STA 2305 / SST 409: Stochastic Processes

Complete Course Notes — Outline, Concepts and Worked Derivations

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This document is built on the thirty-page STA2305 notebook (course outline and key concepts), which supplies the course structure and the order of the topics. Every topic that also appears in the two companion sets — the seven-page "extended notes" set (A) and the fifteen-page "lecture notes on random variables" set (B) — has been merged into that topic's section: the richer derivation is kept and the other variant is noted. Topics that occur only in the outline notebook (the Poisson process solved through difference-differential equations, birth–death processes, Feller's method, queuing models, Markov chains) stand as given there. Appendix C maps every page of all three notebooks to the section that carries it.

Abbreviations as used in the originals: r.v. = random variable; st. pr. = stochastic process; g.f. = generating function; p.g.f. = probability generating function; d.d.e. = difference differential equation(s); m.g.f. = moment generating function; dist./distr. = distribution; mtd. = method; qty = quantity; prob. = probability; nos = numbers; pop^n = population; $o(\Delta t)$ = "little-oh" of Δt.

— outline notebook, p. 1 (dated Wednesday 12th Feb 2025) —

1. Course Outline

Course Outline: To be given later.

1.1 Course description

- Random phenomena in time & space. Bernoulli, Poisson, stochastic discrete random walk, Markov chains: property and discrete time, continuous time.
- Sojourn times & stationary distributions.
- Classification of states.
- Absorption probabilities.
- Expected times of transitions.
- Applications in relevant sectors.
- Random walk, and generating functions.
- Recurrent events.
- Pure birth processes, birth & death processes.
- Branching processes.
- Queuing models: Poisson input, negative exponential and service stability.
- Wiener process.
- Brownian motion process with drift, geometric Brownian motion process.

1.2 Learning outcomes

On completing the course the learner should be able to:

1. Describe random phenomena in time & space.
2. Describe a stochastic process in discrete & continuous time.
3. Describe Markov chains.
4. Classify states and absorption probabilities.
5. Describe random walk and ergodic motion.
6. Explain recurrent events.
7. Describe queuing theory.

2. Introduction: Random Phenomena and Stochastic Processes

2.1 Random phenomena

- A random phenomenon is an empirical situation that obeys probabilistic laws rather than deterministic laws.
- A random phenomenon that arises through a process (for example the motion of a particle in a Brownian motion or the price of a stock in a financial market, etc.) which is developing in time in a manner controlled by probabilistic laws is called a stochastic process.

2.2 Stochastic process / random process

- The theory of stochastic processes deals with systems which develop in time or space in accordance with probabilistic laws.
- The theory is widely used in science and technology; for instance, it can be applied in finance to model the random behaviour that an asset evolves over time.
- The word “stochastic” has a Greek origin meaning “guess”.
- A stochastic process is the counterpart to a deterministic process in probability theory.
- We can therefore define a stochastic process as a sequence of random variables in a given system:
$$\{X_t\}_{t \in T}$$
- Mathematically a stochastic process is a sequence (a collection) of random variables
$$\{X_t\}_{t \in T}, \qquad T \in \mathbb{R}$$
- T is called the **index set** of the process. Note that X_t and T are random variables (and can be discrete or continuous).
- By stochastic processes, we mean a set of r.v. $\{X(t)\}$ depending on a real parameter t which varies in a certain set of natural nos. A stochastic process is denoted by
$$\{X(t),\ t \in T\}.$$
- In many problems of the theory of st. pr., the parameter t means time, though it can denote any other physical qty such as space in which the process occurs.

2.3 Time, states and state space

- If T is discrete, i.e. $T = \{0, 1, 2, \ldots\}$, then the process $\{X_t\}_{t \in T}$ is called a **discrete-time stochastic process**.
- If T is continuous (an interval of $\mathbb{R}$, i.e. $T = [0, \infty)$), then the process $\{X_t\}_{t \in T}$ is called a **continuous-time stochastic process**.
- Given any time t, the possible values of the process $\{X_t\}_{t \in T}$ are called the **states** of the process at time t.
- The set of all the states of a stochastic process is called a **state space**.
- Note that the state space is discrete if each random variable X_t, for the different values of the time t, is a discrete random variable; otherwise it is a continuous state space.

> *“Take note of notation”: Poisson process, Markov process, Wiener process.*

2.4 Examples of stochastic processes

i) Random walk. Where we are concerned with the motion of a particle in a straight line which moves in discrete jumps with certain probabilities, i.e. at each stage it moves one unit to the right or to the left with probability p or $1 - p$. If initially the

particle is at the origin, then its position $X(n)$ after n jumps is a discrete r.v. depending on the discrete parameter n. Thus $\{X(n),\ n \in T\}$ is a stochastic process.

ii) Number of telephone calls. The no. of telephone calls $X(t)$ during the arbitrary period of time t is a discrete r.v. depending on the continuous parameter t. Thus $\{X(t),\ t \in T\}$ is a stochastic process.

iii) Water level of a river. The water level of a river under a certain bridge, measured daily at any moment t, is a continuous r.v. depending on the continuous parameter t. Thus $\{X(t),\ t \in T\}$ is a stochastic process.

iv) Brownian motion. The no. of particles executing a Brownian motion with small, rapid random steps is a discontinuous r.v. depending on the continuous parameter t. Thus $\{X(t),\ t \in T\}$ constitutes a stochastic process.

2.5 Common and important stochastic processes

The following are some of the most common and important stochastic processes:

1. Poisson process
2. Random walk
3. Wiener process
4. Markov process (chains)
5. Renewal process, etc.

— outline notebook, p. 2 —

3. Generating Functions

3.1 Definition

Let $a_0, a_1, a_2, \ldots$ be a sequence of real numbers. Using a variable, say s, we may define a function

$$A(s) = a_0 + a_1 s + a_2 s^2 + \cdots = \sum_{k=0}^{\infty} a_k s^k \qquad \cdots(1)$$

which is a power series.

- If this power series converges in some interval, say $-s_0 < s < s_0$ (the lecture notes write $s_0 < s < s_1$), then equation (1) above is called the **generating function** (g.f.) of the sequence
$$\{a_k\} = \{a_0, a_1, a_2, \ldots\}$$
- The variable s itself has no particular significance.

3.2 Extracting the coefficients by differentiation

Suppose we differentiate the g.f. $A(s)$ k times w.r.t. s, set $s = 0$ and finally divide by $k!$. We obtain the constant a_k, i.e.

$$a_k = \frac{1}{k!}\frac{d^k A(s)}{ds^k}\bigg|_{s=0}$$

For instance, to get a_2 in equation (1) we have:

$$\frac{dA(s)}{ds} = a_1 + 2a_2 s + 3a_3 s^2 + 4a_4 s^3 + \cdots$$

$$\Rightarrow \quad \frac{d^2 A(s)}{ds^2} = 2a_2 + 6a_3 s + 12a_4 s^2 + \cdots$$

$$\Rightarrow \quad \frac{d^2 A(s)}{ds^2}\bigg|_{s=0} = 2a_2$$

$$\Rightarrow \quad \frac{1}{2!}\frac{d^2 A(s)}{ds^2}\bigg|_{s=0} = \frac{2a_2}{2!} = a_2$$

Carrying the same idea one step further (the lecture notes give the full ladder):

$$\left.\frac{dA(s)}{ds}\right|_{s=0} = a_1, \qquad \left.\frac{d^2A(s)}{ds^2}\right|_{s=0} = 2a_2, \qquad \left.\frac{d^3A(s)}{ds^3}\right|_{s=0} = 3!\,a_3, \qquad \left.\frac{d^kA(s)}{ds^k}\right|_{s=0} = k!\,a_k$$

$$\therefore \quad a_2 = \left.\frac{1}{2}\frac{d^2A(s)}{ds^2}\right|_{s=0}, \qquad a_3 = \left.\frac{1}{3!}\frac{d^3A(s)}{ds^3}\right|_{s=0}, \qquad a_k = \left.\frac{1}{k!}\frac{d^kA(s)}{ds^k}\right|_{s=0}$$

3.3 Examples

Example 1

If $a_k = 1$ for all k, i.e. the sequence $\{a_k\} = \{1, 1, 1, \ldots\}$, obtain the g.f. of this sequence.

Soln. By definition the g.f. $A(s)$ is given by

$$A(s) = a_0 + a_1 s + a_2 s^2 + a_3 s^3 + \cdots = 1 + s + s^2 + s^3 + \cdots = \frac{1}{1-s}, \qquad |s| < 1$$

(the lecture notes and extended set state this same result as $a_0 = 1, a_1 = 1, a_2 = 1, \ldots$ and $\sum_{k=0}^{\infty} a_k s^k = 1 + s + s^2 + \cdots$.)

Example 2

Consider the sequence $\{0, 0, 0, 1, 1, 1, \ldots\}$. Obtain the g.f. of $\{a_k\} = \{0, 0, 0, 1, 1, 1, \ldots\}$.

Soln.

$$\begin{aligned} A(s) &= a_0 + a_1 s + a_2 s^2 + a_3 s^3 + a_4 s^4 + \cdots \\ &= s^3 + s^4 + s^5 + \cdots \\ &= s^3\left(1 + s + s^2 + s^3 + \cdots\right) = \frac{s^3}{1-s}, \qquad |s| < 1 \end{aligned}$$

Example 3

Let $\{a_k\} = \left\{\frac{1}{k!}\right\}$ be a sequence of real numbers. Obtain the g.f. of $\{a_k\}$.

Soln.

$$A(s) = a_0 + a_1 s + a_2 s^2 + a_3 s^3 + \cdots = 1 + \frac{s}{1!} + \frac{s^2}{2!} + \frac{s^3}{3!} + \cdots \qquad \text{(Maclaurin series)} = e^s$$

Example 4

The sequence $\{a_k\} = \left\{\frac{n!}{k!\,(n-k)!}\right\}$ for a fixed n. What is the g.f. of $\{a_k\}$?

Soln.

$$A(s) = a_0 + a_1 s + a_2 s^2 + \cdots = \sum_{k=0}^{\infty} a_k s^k = \sum_{k=0}^{n} \binom{n}{k} s^k = (s+1)^n$$

This follows from the binomial expansion

$$(a+b)^n = \sum_{x=0}^{n} \binom{n}{x} a^x b^{n-x}$$

and any value of k greater than n gives zero.

Example 5 (recurrence relation)

This example appears in the extended set and in the lecture notes; the fuller step-by-step version is kept.

Let $a_0, a_1, a_2, \ldots, a_n$ be a sequence of numbers satisfying the recurrence relation

$$a_n + a_{n-1} - 16a_{n-2} + 20a_{n-3} = 0 \qquad \text{for } n \geq 3$$

subject to the initial values $a_0 = 0, a_1 = 1$ & $a_2 = -1$. Obtain the generating function of the sequence $\{a_n\}$.

Soln. Let $A(s) = \sum_{n=0}^{\infty} a_n s^n$ be the g.f. of $\{a_n\}$. Multiplying the recurrence by s^n and summing over n:

$$\sum_{n=3}^{\infty} a_n s^n + \sum_{n=3}^{\infty} a_{n-1} s^n - 16 \sum_{n=3}^{\infty} a_{n-2} s^n + 20 \sum_{n=3}^{\infty} a_{n-3} s^n = 0$$

$$\left(A(s) - a_0 - a_1 s - a_2 s^2\right) + s \sum_{n=3}^{\infty} a_{n-1} s^{n-1} - 16 s^2 \sum_{n=3}^{\infty} a_{n-2} s^{n-2} + 20 s^3 \sum_{n=3}^{\infty} a_{n-3} s^{n-3} = 0$$

$$\left(A(s) - s + s^2\right) + s\left(A(s) - a_0 - a_1 s\right) - 16 s^2\left(A(s) - a_0\right) + 20 s^3 A(s) = 0$$

$$A(s) - s + s^2 + sA(s) - s^2 - 16s^2A(s) + 20s^3A(s) = 0$$

Making $A(s)$ the subject:

$$A(s)\left(1 + s - 16s^2 + 20s^3\right) = s$$

$$\therefore \quad A(s) = \frac{s}{1 + s - 16s^2 + 20s^3} \qquad \text{is the g.f. of } \{a_n\}$$

— outline notebook, pp. 2–3 —

4. Probability Generating Function (p.g.f.)

4.1 Definition

Suppose that X is a random variable which assumes non-negative integral values $0, 1, 2, \ldots$ and that $\Pr(X = k) = p_k$ such that $\sum p_k = 1$. Then the p.g.f., denoted $P(s)$, of the sequence $\{p_k\}$ is given by

$$P(s) = p_0 + p_1 s + p_2 s^2 + p_3 s^3 + \cdots = \sum_{k=0}^{\infty} p_k s^k$$

$$\Rightarrow \quad P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} s^k \Pr(X = k) = E\left[s^X\right]$$

where $E[s^X]$ — written $F(s^x)$ in the lecture notes — is the expectation of the function s^X of the random variable X.

Note that the power series $P(s)$ converges for at least $-1 \le s \le 1$; clearly $P(1) = 1$.

4.2 A sequence of probabilities that is not a p.g.f.

- Note that it is possible to have a sequence of probabilities which does not form a p.g.f., as long as the sum of the probabilities is not equal to one.

e.g. If $p_0 = 0, p_1 = \frac{1}{4}, p_2 = \frac{1}{2}, p_3 = \frac{1}{3}, p_4 = \frac{3}{4}$, then

$$A(s) = p_0 + p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 = \frac{1}{4}s + \frac{1}{2}s^2 + \frac{1}{3}s^3 + \frac{3}{4}s^4$$

$A(s)$ is a g.f. of $\{p_k\}$ but not a p.g.f., since

$$p_0 + p_1 + p_2 + p_3 + p_4 \neq 1.$$

4.3 Determination of the mean and variance using the p.g.f.

The p.g.f. is

$$P(s) = \sum_{k=0}^{\infty} p_k s^k$$

Differentiating once:

$$P'(s) = \sum_{k=0}^{\infty} k\, p_k s^{k-1}, \qquad -1 < s < 1$$

$$\Rightarrow \quad P'(s)\Big|_{s=1} = P'(1) = \sum_{k=0}^{\infty} k\, p_k$$

Thus the expected value of X is

$$E[x] = \sum_{k=0}^{\infty} k\, p_k = P'(1)$$

To get the mean, differentiate w.r.t. s and set $s = 1$ (first derivative).

Differentiating twice:

$$P''(s) = \sum_{k=0}^{\infty} k(k-1)\, p_k s^{k-2}$$

$$\Rightarrow \quad P''(s)\Big|_{s=1} = P''(1) = \sum_{k=0}^{\infty} k(k-1)\, p_k$$

Again

$$E\big[x(x-1)\big] = \sum_{k=0}^{\infty} k(k-1)p_k = P''(1)$$

$$\& \qquad E\big[x^2\big] = E\big[x(x-1)+x\big] = E\big[x(x-1)\big] + E[x] = P''(1) + P'(1)$$

Therefore

$$\mathrm{Var}(x) = E(x^2) - \big(E(x)\big)^2 = P''(1) + P'(1) - \Big(P'(1)\Big)^2$$

4.4 Further properties of the p.g.f.

(a) Link with the moment generating function

When $s = e^t$ then

$$P(s) = P\left(e^t\right) = E\left(e^{tx}\right)$$

which is the m.g.f. of x.

(b) The integral of the p.g.f.

$$\int_0^1 P(s)\,ds = \int_0^1 \left(\sum_{k=0}^{\infty} p_k s^k\right) ds = \int_0^1 \left(p_0 + p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 + \cdots\right) ds$$

$$= \int_0^1 p_0\,ds + \int_0^1 p_1 s\,ds + \int_0^1 p_2 s^2\,ds + \int_0^1 p_3 s^3\,ds + \int_0^1 p_4 s^4\,ds + \cdots$$

$$= p_0 + \frac{p_1}{2} + \frac{p_2}{3} + \frac{p_3}{4} + \frac{p_4}{5} + \frac{p_5}{6} + \cdots = \sum_{k=0}^{\infty} \frac{p_k}{k+1} = E\left(\frac{1}{X+1}\right)$$

(c) Example: the p.g.f. of a linear transform $Y = mX + n$

Let X be a r.v. with p.g.f. $P(s)$. Find the p.g.f. of $Y = mX + n$, where m, n are integers & $m \neq 0$.

Let $P_Y(s)$ be the p.g.f. of Y. Then

$$P_Y(s) = E\left(s^Y\right) = E\big[s^{mX+n}\big] = E\big[s^{mX}s^n\big] = s^n E\Big[\left(s^m\right)^X\Big] = s^n P\left(s^m\right)$$

(with $P_x(s) = E(s^x)$ as noted in the margin.)

(d) Recovering the probabilities from the p.g.f.

Remark i) We can obtain the probabilities p_k from the p.g.f.; p_k can be found from $P(s)$, i.e.

$$p_k = \frac{1}{k!}\left[\frac{d^k P(s)}{ds^k}\bigg|_{s=0}\right]$$

Remark ii) p_k is also given by the coefficient of s^k in the expansion of $P(s)$ as a power series in s, e.g.

If

$$P(s) = e^{-\lambda(1-s)} = e^{-\lambda}e^{\lambda s} = e^{-\lambda}\left[1 + \frac{\lambda s}{1!} + \frac{(\lambda s)^2}{2!} + \cdots + \frac{(\lambda s)^k}{k!} + \cdots\right]$$

then

$$p_k = \text{coefficient of } s^k = e^{-\lambda}\frac{\lambda^k}{k!}$$

which is a Poisson distribution with parameter λ.

5. The p.g.f. and Moments of Standard Distributions

5.1 Bernoulli distribution

Let X have a Bernoulli distribution with parameter p, so that

$$\Pr(X = k) = p_k = p^k q^{1-k}, \qquad k = 0, 1, \qquad q = 1 - p$$

Then the p.g.f. of X is

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{1} p^k q^{1-k} s^k = \sum_{k=0}^{1} (ps)^k q^{1-k} = q + ps$$

Mean of X: since $P(s) = q + ps$, we have $P'(s) = p$, so

$$E(x) = P'(1) = p$$

Variance of X:

$$\text{Var} = P''(1) + P'(1) - \left(P'(1)\right)^2$$

But if $P'(s) = p$ then $P''(s) = 0$ and $P''(1) = 0$. So

$$\text{Var}(x) = 0 + p - p^2 = p(1-p) = pq$$

since $p + q = 1$.

5.2 Binomial distribution

Let X have a Binomial distribution with parameters n and p, so that

$$\Pr(X = k) = p_k = \binom{n}{k} p^k q^{n-k}, \qquad k = 0, 1, 2, \ldots, n, \qquad q + p = 1$$

Then the p.g.f. of X is

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{n} \binom{n}{k} p^k q^{n-k} s^k = \sum_{k=0}^{n} \binom{n}{k} (ps)^k q^{n-k} = (ps + q)^n$$

and this follows from the binomial expansion.

Mean of X: $E(X) = P'(1)$. But

$$P'(s) = np\,(ps + q)^{n-1}$$

$$\Rightarrow \quad E(X) = P'(1) = np(p + q)^{n-1} = np \qquad \text{since } p + q = 1$$

Variance of X:

$$\text{Var}(X) = P''(1) + P'(1) - \left(P'(1)\right)^2 = P''(1) + np - (np)^2$$

But

$$P''(s) = np(n-1)p\,(ps + q)^{n-2}$$

$$\Rightarrow \quad P''(1) = np(n-1)p\,(p + q)^{n-2} = n(n-1)p^2 = n^2p^2 - np^2$$

So

$$\text{Var}(X) = n^2p^2 - np^2 + np - (np)^2 = np - np^2 = np(1-p) = npq$$

5.3 Poisson distribution

$$\Pr(X = k) = p_k = e^{-\lambda}\frac{\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

The p.g.f. of X is

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} e^{-\lambda}\frac{\lambda^k}{k!} s^k = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda s)^k}{k!}$$

$$= e^{-\lambda}\left[1 + \lambda s + \frac{(\lambda s)^2}{2!} + \frac{(\lambda s)^3}{3!} + \cdots\right] = e^{-\lambda} e^{\lambda s} = e^{-\lambda(1-s)}$$

Exercise (outline notebook): use the above p.g.f. to show that $E(x) = \text{Var}(x) = \lambda$.

Solution (worked out in the extended set):

$$P'(s) = \lambda e^{-\lambda(1-s)} \qquad P''(s) = \lambda^2 e^{-\lambda(1-s)}$$

$$E(x) = P'(1) = \lambda e^{-\lambda(1-1)} = \lambda e^0 = \lambda \qquad P''(1) = \lambda^2 e^{-\lambda(1-1)} = \lambda^2 e^0 = \lambda^2$$

$$\text{Var}(x) = P''(1) + P'(1) - \left(P'(1)\right)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

5.4 Geometric distribution

Exercise (outline notebook): Let X have a geometric distribution of the form

$$\Pr(X = k) = p_k = pq^k, \qquad k = 0, 1, 2, 3, \ldots, \qquad p + q = 1.$$

Obtain the p.g.f. of X and hence use the p.g.f. to obtain the mean and variance of X.

Solution (extended set and lecture notes):

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} p\, q^k s^k = p \sum_{k=0}^{\infty} (qs)^k$$

$$= p\left(1 + qs + (qs)^2 + (qs)^3 + \cdots\right) = p\left(\frac{1}{1-qs}\right) = \frac{p}{1-qs}$$

Writing $u = p$ and $v = 1 - qs$ and using the quotient rule,

$$P'(s) = \frac{u'v - uv'}{v^2} = \frac{p(1-qs) - p(-q)s}{(1-qs)^2} = \frac{p - pqs + pqs}{(1-qs)^2} = \frac{pq}{(1-qs)^2}$$

$$P''(s) = \frac{2pq^2}{(1-qs)^3}$$

$$E(x) = P'(1) = \frac{pq}{(1-q)^2} = \frac{pq}{p^2} = \frac{q}{p}$$

$$P''(1) = \frac{2pq^2}{(1-q)^3} = \frac{2pq^2}{p^3} = \frac{2q^2}{p^2}$$

$$\mathrm{Var}(x) = P''(1) + P'(1) - \left(P'(1)\right)^2 = \frac{2q^2}{p^2} + \frac{q}{p} - \frac{q^2}{p^2}$$

$$= \frac{q^2}{p^2} + \frac{q}{p} = \frac{q^2 + pq}{p^2} = \frac{q(q+p)}{p^2} = \frac{q}{p^2}, \qquad (p+q) = 1$$

Note on the outline notebook's own working (p. 16): it expands the series from $k = 1$, i.e. $P(s) = ps + pqs^2 + pq^2s^3 + \cdots = \frac{ps}{1-qs}$, which is the same distribution shifted by one success. With that shift $E(x) = \frac{p}{(1-q)^2} = \frac{1}{p}$, while the variance is unchanged, $\frac{2q}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{q}{p^2} = \frac{q}{(1-q)^2}$.

— outline notebook, p. 4 —

6. Tail or Cumulative Probabilities

6.1 Upper tail probabilities

Suppose X has a probability distribution

$$\Pr(X = k) = p_k, \qquad k = 0, 1, 2, \ldots$$

with p.g.f. $P(s) = \sum_{k=0}^{\infty} p_k s^k$. We define the **upper tail probability**

$$q_k = \Pr(X > k) = p_{k+1} + p_{k+2} + \cdots$$

with generating function

$$Q(s) = \sum_{k=0}^{\infty} q_k s^k \qquad \left(\text{since } \sum_{k=0}^{\infty} q_k \neq 1, \text{ this is a g.f. and not a p.g.f.}\right)$$

We want to get the g.f. of the upper tail probabilities in terms of the p.g.f. of X. That is: if

$$q_k = p_{k+1} + p_{k+2} + p_{k+3} + \cdots$$
$$\Rightarrow \quad q_{k-1} = p_k + p_{k+1} + p_{k+2} + \cdots$$
$$\Rightarrow \quad q_{k-1} - q_k = p_k, \qquad k \geq 1$$
$$\Rightarrow \quad p_k = q_{k-1} - q_k \qquad \cdots ①$$

But

$$P(s) = \sum_{k=0}^{\infty} p_k s^k \qquad \text{and} \qquad Q(s) = \sum_{k=0}^{\infty} q_k s^k$$

We introduce s^k in equation ① and sum over k to get

$$\sum_{k=1}^{\infty} p_k s^k = \sum_{k=1}^{\infty} q_{k-1} s^k - \sum_{k=1}^{\infty} q_k s^k$$

$$\Rightarrow \quad P(s) - p_0 = s \sum_{k=1}^{\infty} q_{k-1} s^{k-1} - \left[Q(s) - q_0\right]$$

$$\Rightarrow \quad P(s) - p_0 = sQ(s) - Q(s) + q_0$$

Making $Q(s)$ the subject of the formula,

$$Q(s)(s-1) = P(s) - (p_0 + q_0)$$

Note that

$$q_k = p_{k+1} + p_{k+2} + p_{k+3} + \cdots \qquad \cdots ①$$
$$q_0 = p_1 + p_2 + p_3 + \cdots$$
$$p_0 + q_0 = p_0 + p_1 + p_2 + p_3 + \cdots = 1 \qquad \text{i.e. } \sum p_k = 1$$
$$\frac{Q(s)(s-1)}{(s-1)} = \frac{P(s)-1}{(s-1)} \qquad \text{since } p_0 + q_0 = 1$$
$$\Rightarrow \quad Q(s) = \frac{P(s)-1}{s-1} = \frac{1-P(s)}{1-s}, \qquad |s| < 1$$

Alternative method (lecture notes): direct summation of the tail

$$\phi(s) = \sum_{k=0}^{\infty} q_k s^k = q_0 + q_1 s + q_2 s^2 + q_3 s^3 + \cdots$$
$$= (p_1 + p_2 + p_3 + \cdots) + (p_2 + p_3 + p_4 + \cdots)s + (p_3 + p_4 + p_5 + \cdots)s^2 + (p_4 + p_5 + p_6 + \cdots)s^3 + \cdots$$
$$= p_1 + p_2(1+s) + p_3\left(1 + s + s^2\right) + p_4\left(1 + s + s^2 + s^3\right) + \cdots$$
$$= p_1 + p_2(1+s) + p_3\frac{\left(1-s^3\right)}{(1-s)} + p_4\frac{\left(1-s^4\right)}{(1-s)} + \cdots$$
$$= \frac{p_1(1-s) + p_2\left(1-s^2\right) + p_3\left(1-s^3\right) + p_4\left(1-s^4\right) + \cdots}{(1-s)}$$
$$= \frac{(p_1 + p_2 + p_3 + p_4 + \cdots) - \left(p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 + \cdots\right)}{(1-s)}$$
$$= \frac{(p_0 + p_1 + p_2 + p_3 + p_4 + \cdots) - \left(p_0 + p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4 + \cdots\right)}{(1-s)}$$
$$\therefore \quad \phi(s) = \frac{1-P(s)}{1-s}, \qquad -1 < s < 1$$

which is the same result as $Q(s)$ above.

6.2 Lower tail probabilities

For the lower tail probability, let

$$q_k = \Pr(X \le k) = p_k + p_{k-1} + p_{k-2} + \cdots + p_1 + p_0, \qquad k = 0, 1, 2, \ldots$$

so that

$$q_{k-1} = p_{k-1} + p_{k-2} + p_{k-3} + \cdots + p_1 + p_0$$
$$\Rightarrow \quad q_k - q_{k-1} = p_k, \qquad k \ge 1$$
$$\Rightarrow \quad p_k = q_k - q_{k-1} \qquad \cdots ②$$

But

$$P(s) = \sum_{k=0}^{\infty} p_k s^k \qquad \text{and} \qquad Q(s) = \sum_{k=0}^{\infty} q_k s^k$$

From equation ② we introduce s^k and sum over k to get

$$\sum_{k=1}^{\infty} p_k s^k = \sum_{k=1}^{\infty} q_k s^k - \sum_{k=1}^{\infty} q_{k-1} s^k$$
$$\Rightarrow \quad P(s) - p_0 = Q(s) - q_0 - s\sum_{k=1}^{\infty} q_{k-1} s^{k-1}$$
$$\Rightarrow \quad P(s) - p_0 = Q(s) - q_0 - sQ(s)$$

— outline notebook, p. 5 —

$$\Rightarrow \quad Q(s)(1-s) = P(s) - p_0 + q_0$$
$$\Rightarrow \quad Q(s) = \frac{P(s)-0}{1-s} \qquad \text{since } p_0 = q_0$$
$$\Rightarrow \quad Q(s) = \frac{P(s)}{1-s}, \qquad |s| < 1$$

7. Probability Generating Function of the Sum of Independent Random Variables

Here we consider two cases: **fixed numbers** and **random numbers**.

7.1 Case one: a fixed number of terms

We can get the p.g.f. of the sum of independent random variables, i.e.

$$S_n = X_1 + X_2 + \cdots + X_n, \qquad n \text{ fixed},$$

using two methods: (1) using convolution, and (2) using expectation. The expectation method is carried out in the notebooks as follows.

Let X and Y be two independent random variables, and let $A(s)$, $B(s)$ and $C(s)$ be the p.g.f.s of X, Y and Z respectively, where $Z = X + Y$. Then

$$C(s) = E\left[s^Z\right] = E\left[s^{X+Y}\right] = E\left[s^X\right] \cdot E\left[s^Y\right] \qquad \text{due to independence}$$
$$\therefore \quad C(s) = A(s) \cdot B(s)$$

This means that the p.g.f. of the sum of two independent random variables (X and Y) is simply the product of the p.g.f.s of X and Y (the lecture notes state the same result as $P_z(s) = P_x(s)P_y(s)$ for the non-negative integral variables $a_k = P(X = k)$ and $b_j = P(Y = j)$).

This result can be extended to the case where we have n independent random variables $X_1, X_2, \ldots, X_n$: the p.g.f. of the sum $S_n = X_1 + X_2 + \cdots + X_n$ is the product of the p.g.f.s of the individual random variables:

$$P_{S_n}(s) = E\left[s^{S_n}\right] = E\left[s^{x_1 + x_2 + \cdots + x_n}\right] = E\left(s^{x_1}\right) E\left(s^{x_2}\right) \cdots E\left(s^{x_n}\right)$$
$$= P_{x_1}(s)\, P_{x_2}(s) \cdots P_{x_n}(s)$$

If the random variables $X_1, X_2, \ldots, X_n$ are also identically distributed, each with p.g.f. $P(s)$, then the p.g.f. of S_n becomes

$$C(s) = P_1(s) \cdot P_2(s) \cdot P_3(s) \cdots P_n(s) = P(s) \cdot P(s) \cdots P(s) = \left[P(s)\right]^n$$

Example one: the sum of two Poisson variables

Let X_1 and X_2 be two independent Poisson random variables with parameters λ_1 and λ_2 respectively. Find the distribution of the random variable $Z = X_1 + X_2$.

Soln. Let $P_1(s)$ and $P_2(s)$ be the p.g.f.s of X_1 and X_2 respectively.

$$P_1(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} \left(e^{-\lambda_1} \frac{\lambda_1^k}{k!} \right) s^k = e^{-\lambda_1(1-s)}$$
$$P_2(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} \left(e^{-\lambda_2} \frac{\lambda_2^k}{k!} \right) s^k = e^{-\lambda_2(1-s)}$$

Therefore the p.g.f. of Z, denoted by $P_z(s)$, is given by

$$P_z(s) = P_1(s) \cdot P_2(s) = \left(e^{-\lambda_1(1-s)} \right) \left(e^{-\lambda_2(1-s)} \right) = e^{-(\lambda_1 + \lambda_2)(1-s)}$$

which is the p.g.f. of another version of a Poisson distribution with parameter $(\lambda_1 + \lambda_2)$. So the distribution of Z is

$$\Pr(Z = k) = e^{-(\lambda_1 + \lambda_2)} \frac{(\lambda_1 + \lambda_2)^k}{k!}, \qquad k = 0, 1, 2, 3, \ldots$$
$$= 0, \quad \text{elsewhere}$$

The result extends to n independent Poisson variables: the sum $S_n = X_1 + X_2 + \cdots + X_n$ is Poisson with mean

$$\sum_{i=1}^{n} \lambda_i$$

where λ_i is the mean of x_i.

Example two: the sum of Bernoulli variables

Let X be a Bernoulli random variable with success parameter p. Let the Bernoulli trials be $X_1, X_2, \ldots, X_n$. Find the distribution of the random variable $Y = X_1 + X_2 + \cdots + X_n$.

Soln. Let $P_1(s), P_2(s), \ldots, P_n(s)$ be the p.g.f.s of $X_1, X_2, \ldots, X_n$ and $P_Y(s)$ be the p.g.f. of Y. Then

$$P_Y(s) = P_1(s) \cdot P_2(s) \cdots P_n(s)$$
$$= P(s) \cdot P(s) \cdots P(s) \qquad \text{since the variables are identical}$$
$$= (q + ps)(q + ps) \cdots (q + ps) = (q + ps)^n$$

which is the p.g.f. of a binomial random variable. So the distribution of Y is

$$\Pr(Y = k) = \binom{n}{k} p^k q^{n-k}, \qquad k = 0, 1, 2, \ldots, n$$
$$= 0, \quad \text{elsewhere}$$

7.2 Case two: a random number of terms

So far we have considered the sum $S_n = X_1 + X_2 + \cdots + X_n$ of a fixed number of mutually independent random variables. Cases exist where we have to consider the sum of a random number N of random variables.

Illustration. Suppose X_i denotes the no. of people involved in the ith accident in a day in some place, and if the number N of accidents happening in a day is a random variable, then the sum $S_N = X_1 + X_2 + \cdots + X_N$ denotes the total no. of persons involved in accidents in a day.

The following theorem gives the p.g.f. of the sum S_N.

Theorem

Let x_i, $i = 1, 2, \ldots, N$ be i.i.d. (independent and identically distributed) random variables with the probability

$$\Pr(x_i = k) = p_k$$

and the corresponding p.g.f.

$$P(s) = \sum_{k=0}^{\infty} p_k s^k, \qquad k = 0, 1, 2, \ldots$$

Let $S_N = x_1 + x_2 + \cdots + x_N$ where N is a random variable independent of the x_i's. Let the distribution of N be given by

$$\Pr(N = n) = g_n$$

and the corresponding p.g.f. of N be

$$G(s) = \sum_{n=0}^{\infty} g_n s^n.$$

Then the p.g.f. of S_N, denoted by $H(s)$, is given by the corresponding compounding function $G\big(P(s)\big)$, i.e. the p.g.f. of S_N is

$$H(s) = \sum_{j=0}^{\infty} \Pr(S_N = j)\, s^j = G\big(P(s)\big)$$

Mean of S_N

$$E[S_N] = H'(s)\big|_{s=1} = H'(1)$$

But $H(s) = G\big(P(s)\big)$, so

$$\Rightarrow \quad H'(s) = P'(s) \cdot G'\big(P(s)\big)$$
$$E[S_N] = H'(1) = P'(1) \cdot G'\big(P(1)\big) = E(x_i)\, G'(1) = E(x_i) \cdot E(N)$$

(the lecture notes write the same chain as
$H'(1) = G'[P(1)]P'(1) = G'(1)P'(1) = E(N)E(x_i)$.*)*

Exercise. Use the p.g.f. technique to express the variance of S_N in terms of the means and variances of X_i and N.

Hint: $\text{Var}(S_N) = H''(1) + H'(1) - \big(H'(1)\big)^2$.

Example one: a Poisson number of terms

Suppose N has a Poisson distribution with mean λ and x_i, $i = 1, 2, \ldots, N$ are i.i.d. random variables with p.g.f. $P(s)$. Find the p.g.f. of $S_N = x_1 + x_2 + \cdots + x_N$.

Soln. The p.g.f. of S_N is $H(s) = G\big(P(s)\big)$.

For a Poisson random variable N with mean λ, its p.g.f. is

$$P(s) = \sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} s^k = e^{-\lambda(1-s)}$$

Thus the p.g.f. of S_N is

$$H(s) = G\big(P(s)\big) = e^{-\lambda\left(1 - P(s)\right)}$$

We can also get the mean of S_N, i.e.

$$E\big[S_N\big] = E(X_i) \cdot E(N) = \lambda\, E(x_i)$$

- Note that the distribution having a g.f. of the form $e^{-\lambda(1 - P(s))}$, where $P(s)$ is itself a generating function, is called a **compound Poisson distribution**.

— outline notebook, p. 6 —

8. Stochastic Processes with Discrete States in Continuous Time

Here we consider a class of stochastic processes in continuous time and with a discrete state space. One of the simplest of these processes is the famous Poisson process.

The Poisson process serves as a mathematical model for a wide range of empirical phenomena, e.g. we use the Poisson process to model the following situations:

- the arrival of calls at a telephone exchange;
- emission of particles from a radioactive source, etc.;
- no. of car accidents in a city within a given time interval.

8.1 Definition of the Poisson process

Consider events occurring singly in time. Let $N(t, t + \Delta t)$ denote the no. of events in the time interval $(t, t + \Delta t]$.

Suppose that for some positive constant λ, as $\Delta t \to 0$,

$$\Pr\big\{N(t, t + \Delta t) = 0\big\} = 1 - \lambda\Delta t + o(\Delta t) \qquad \cdots ①$$
$$\Pr\big\{N(t, t + \Delta t) = 1\big\} = \lambda\Delta t + o(\Delta t) \qquad \cdots ②$$

so that

$$\Pr\big\{N(t, t + \Delta t) > 1\big\} = o(\Delta t) \qquad \cdots ③$$

- Furthermore, $N(t, t + \Delta t)$ is completely independent of occurrences in the time interval $(0, t]$, where $o(\Delta t)$ denotes a function tending to 0 faster than Δt; by definition $o(\Delta t)$ denotes a function of Δt such that

$$\lim_{\Delta t \to 0} \frac{o(\Delta t)}{\Delta t} = 0.$$

Then we say that the stochastic process of point events satisfying the above conditions is a **Poisson process of rate** λ.

8.2 The difference differential equations of the process

In what follows we illustrate a general technique of analysing processes in continuous time with discrete state space, i.e. let $N(t)$ be the no. of events in an interval of fixed length, say $(0, t]$.

Let

$$P_n(t) = \Pr\big(N(t) = n\big)$$

i.e. the probability that there are n events at time t is equal to n.

We develop the differential equations for the sequence of probabilities $\{P_n(t)\}$ as follows: for $\Delta t > 0$,

$$P_n(t + \Delta t) = \Pr\big\{N(t + \Delta t) = n\big\}$$
$$= \Pr\big\{N(t) = n \text{ and } N(t, t + \Delta t) = 0 \quad \text{or} \quad N(t) = n - 1 \text{ and } N(t, t + \Delta t) = 1$$
$$\text{or} \quad N(t) = n - L \text{ and } N(t, t + \Delta t) = L; \quad L = 2, 3, \ldots\big\}$$

— outline notebook, p. 7 —

[in probability, "or" = +; and $N(t)$ and $N(t, t + \Delta t)$ are not independent]

Therefore the above becomes

$$\Rightarrow \quad \Pr\{N(t) = n\} \cdot \Pr\big\{N(t, t + \Delta t) = 0 \mid N(t) = n\big\}$$

$$+\Pr\{N(t)=n-1\}\cdot\Pr\left\{N(t,t+\Delta t)=1 \mid N(t)=n-1\right\}$$
$$+\sum_{L\geq 2}\Pr\{N(t)=n-L\}\cdot\Pr\left\{N(t,t+\Delta t)=L \mid N(t)=n-L\right\}$$

NB: The probabilities corresponding to negative states are by definition taken to be zero.

For the Poisson process the conditional probabilities are constant and are given by equations ①②③ above in the definition of the Poisson process. Now as $\Delta t \to 0$,

$$P_n(t+\Delta t)=P_n(t)\cdot\left[1-\lambda\Delta t+o(\Delta t)\right]+P_{n-1}(t)\left[\lambda\Delta t+o(\Delta t)\right]+o(\Delta t)$$
$$\Rightarrow \quad P_n(t+\Delta t)=P_n(t)-P_n(t)\lambda\Delta t+P_{n-1}(t)\lambda\Delta t+o(\Delta t)$$

This leads to the differential

$$P_n'(t)=\frac{d}{dt}P_n(t)=\lim_{\Delta t\to 0}\frac{P_n(t+\Delta t)-P_n(t)}{\Delta t}$$
$$=\lim_{\Delta t\to 0}\frac{P_n(t)-P_n(t)\lambda\Delta t+P_{n-1}(t)\lambda\Delta t+o(\Delta t)-P_n(t)}{\Delta t}$$
$$=\lim_{\Delta t\to 0}\left\{-\lambda P_n(t)+\lambda P_{n-1}(t)+\frac{o(\Delta t)}{\Delta t}\right\}$$
$$\Rightarrow \quad P_n'(t)=-\lambda P_n(t)+\lambda P_{n-1}(t), \qquad n=1,2,3,\ldots \qquad (X)$$

since $\lim_{\Delta t\to 0}\frac{o(\Delta t)}{\Delta t}=0$, and

$$P_0'(t)=-\lambda P_0(t), \qquad n=0$$

since $P_{-1}(t)=0$.

— outline notebook, p. 8 —

- Equation (X) is a difference differential equation for a Poisson process. What remains is to solve these differential equations, subject to some initial condition, which are always available.
- There are several methods of solving these equations; two of these methods are:
  1. Recursive method.
  2. Probability generating function technique together with the o.p.d. (ordinary partial differential) equations / Lagrange's linear equations technique.

8.3 Lagrange's linear partial differential equation (recap)

Suppose we have a linear partial differential equation (Lagrange's linear equation)

$$P\frac{\partial z}{\partial x}+Q\frac{\partial z}{\partial y}=R \qquad \cdots④$$

subject to some boundary conditions, where P, Q and R are functions of x, y and z. Then

$$\frac{\partial x}{P}=\frac{\partial y}{Q}=\frac{\partial z}{R} \qquad \cdots⑤$$

is known as the auxiliary (subsidiary) set of equation ④.

- From equation ⑤ we have the following equations:
$$\frac{\partial x}{P}=\frac{\partial y}{Q} \qquad \text{and} \qquad \frac{\partial y}{Q}=\frac{\partial z}{R}$$
- Let $u(x,y,z)=$ constant and $v(x,y,z)=$ constant be any two solutions of the auxiliary equations.
- The general solution to equation ④ can be expressed as
$$\Phi(u,v)=0 \qquad \text{or} \qquad u=\phi(v)$$

8.4 Solution of the Poisson process by the p.g.f. technique

Back to the difference differential equation for the Poisson process; we want to solve

$$P_n'(t)=-\lambda P_n(t)+\lambda P_{n-1}(t), \qquad n\geq 1 \qquad \cdots⑦$$

and

$$P_0'(t)=-\lambda P_0(t), \qquad n=0 \qquad \cdots⑧$$

subject to some boundary condition:

$$P_n(0)=1 \text{ for } n=0 \qquad \text{and} \qquad P_n(0)=0 \text{ for } n\neq 0.$$

We use the p.g.f. technique to get the value with the ordinary partial differential (o.p.d.), i.e. define the p.g.f. for the sequence of probabilities $\{P_n(t)\}$ as

$$G(s,t) = \sum_{n=0}^{\infty} P_n(t)\, s^n$$

so that partial differentiation of $G(s,t)$ gives

$$\frac{\partial}{\partial t}G(s,t) = \sum_{n=0}^{\infty} P_n^{'}(t)\, s^n \qquad \text{and} \qquad \frac{\partial}{\partial s}G(s,t) = \sum_{n=0}^{\infty} nP_n(t)\, s^{n-1}$$

From the first set of difference differential equations ⑦ we introduce s^n and sum over n to get

$$\sum_{n=1}^{\infty} P_n^{'}(t)s^n = -\lambda \sum_{n=1}^{\infty} P_n(t)s^n + \lambda \sum_{n=1}^{\infty} P_{n-1}(t)s^n$$

$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) - P_0^{'}(t) = -\lambda\big[G(s,t) - P_0(t)\big] + \lambda s \sum_{n=1}^{\infty} P_{n-1}(t)s^{n-1}$$

$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) - P_0^{'}(t) = -\lambda G(s,t) + \lambda P_0(t) + \lambda s G(s,t)$$

But from the second set of difference differential equations

$$P_0^{'}(t) = -\lambda P_0(t)$$

$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) = -\lambda(1-s)G(s,t) \qquad \cdots ⑨$$

The Lagrange's linear equation becomes

$$1 \cdot \frac{\partial}{\partial t}G(s,t) + 0 \cdot \frac{\partial}{\partial s}G(s,t) = -\lambda(1-s)G(s,t)$$

$\Rightarrow$ The corresponding auxiliary set of equations are

— *outline notebook, p. 9* —

$$\frac{dt}{1} = \frac{ds}{0} = \frac{dG(s,t)}{-\lambda(1-s)G(s,t)}$$

Equating

$$\frac{dt}{1} = \frac{dG(s,t)}{-\lambda(1-s)G(s,t)}$$

$$\int dt = \int \frac{dG(s,t)}{-\lambda(1-s)G(s,t)}$$

$$\Rightarrow \quad \int -\lambda(1-s)\, dt = \int \frac{dG(s,t)}{G(s,t)}$$

$$\Rightarrow \quad -\lambda(1-s)t + c = \ln G(s,t)$$

$$\Rightarrow \quad G(s,t) = e^{-\lambda(1-s)t + c} \qquad \cdots ⑩$$

Using the initial conditions given, i.e.

$$P_n(0) = 1, \quad n = 0 \qquad \text{and} \qquad P_n(0) = 0, \quad n \neq 0,$$

$$\Rightarrow \quad G(s,0) = e^c \qquad \cdots ⑪$$

But by definition of the p.g.f.,

$$G(s,t) = \sum_{n=0}^{\infty} P_n(t)s^n = P_0(t) + P_1(t)s + P_2(t)s^2 + \cdots$$

$$\Rightarrow \quad G(s,0) = P_0(0) + P_1(0)s + P_2(0)s^2 + \cdots = P_0(0) = 1 \qquad \cdots ⑤$$

Equating ⑪ and ⑤ we get

$$G(s,0) = e^c = 1 \quad \Rightarrow \quad c = 0$$

From equation ⑩,

$$G(s,t) = e^{-\lambda(1-s)t}$$

The solution of the above difference differential equations is denoted by $P_n(t)$, which is actually the coefficient of s^n in the $G(s,t)$ expansion.

But

$$G(s,t) = e^{-\lambda(1-s)t} = e^{-\lambda t} \cdot e^{\lambda t s}$$

$$= e^{-\lambda t}\left[1 + \lambda ts + \frac{(\lambda ts)^2}{2!} + \frac{(\lambda ts)^3}{3!} + \cdots + \frac{(\lambda ts)^{n-1}}{(n-1)!} + \frac{(\lambda ts)^n}{n!} + \cdots\right]$$

Thus the solution of the process is

$$P_n(t) = e^{-\lambda t}\frac{(\lambda t)^n}{n!}, \qquad n = 0, 1, 2, \ldots$$
$$= 0, \quad \text{elsewhere}$$

which is a Poisson probability distribution with parameter λt.

8.5 Mean and variance of the number of events

The mean number of events is

$$E(N) = G'(s,t)\big|_{s=1} = G'(1,t)$$

But $G(s,t) = e^{-\lambda(1-s)t}$, so

$$\Rightarrow \quad G'(s,t) = \lambda t\, e^{-\lambda(1-s)t}$$
$$\Rightarrow \quad E(N) = G'(1,t) = \lambda t\, e^{0} = \lambda t$$

and

$$\mathrm{Var}(N) = G''(1,t) + G'(1,t) - \Big(G'(1,t)\Big)^2 = G''(1,t) + \lambda t - (\lambda t)^2$$

But

$$G''(s,t) = \lambda t \cdot \lambda t \cdot e^{-\lambda(1-s)t} \quad \Rightarrow \quad G''(1,t) = \lambda t \cdot \lambda t\, e^{0} = (\lambda t)^2$$
$$\Rightarrow \quad \mathrm{Var}(N) = (\lambda t)^2 + \lambda t - (\lambda t)^2 = \lambda t$$

Exercise. Suppose that the initial conditions for the above Poisson process are

$$P_n(0) = 1 \text{ for } n = i \qquad \text{and} \qquad P_n(0) = 0 \text{ for } n \neq i.$$

Obtain the solution of the process.

— outline notebook, p. 10 —

9. Birth–Death Processes

Here we consider a class of exponential models called birth–death processes, where the Poisson process is treated as a special case.

- Finding the exact analytic formulas for the probabilities of these models involves solving a system of differential equations. We obtain these differential equations in terms of the parameters (i.e. birth and death rates) of the models.
- Define the state $X(t)$ of the system at time t to be the total no. of individuals in the population at that moment.
- Here $N(t)$ is a random variable and we would like to determine its properties by analytic methods (e.g. trying to know what is the expected population size at time t, or what is the population behaviour as time $t \to \infty$).

9.1 Definition

The continuous-time stochastic process $\{N(t), t \geq 0\}$ is a **birth–death process** if there are sequences of non-negative birth rates λ_n, $n \geq 0$ and death rates μ_n, $n \geq 1$, such that the system evolves in time as follows:

i. The elapsed (sojourn) time between entering state n and leaving state n is an exponential random variable with rate $\lambda_n + \mu_n$.

ii. Each transition from state n to state $n+1$ is determined by a Bernoulli random variable with probability

$$p_n = \frac{\lambda_n}{\lambda_n + \mu_n}$$

of jumping to state $n+1$, and with the complementary probability

$$q_n = \frac{\mu_n}{\lambda_n + \mu_n}$$

of jumping to state $n-1$.

- Furthermore, all the elapsed times in (i) and the Bernoulli random variables in (ii) are assumed to be mutually independent.

9.2 The difference differential equations of a birth–death process

Let $N(t)$ be the population size at time t and $P_n(t)$ be the probability of size n at time t, thus

$$P_n(t) = \Pr\{N(t) = n\}$$

We make the following assumptions for the time interval Δt:

① The probability that from a population of size n, a birth occurs within the time interval Δt is

$$\lambda_n \Delta t + o(\Delta t)$$

② The probability that from a population of size n, a death occurs within the time interval Δt is

$$\mu_n \Delta t + o(\Delta t)$$

③ The probability that there is no change in the population size within the time interval Δt is

$$1 - (\lambda_n + \mu_n)\Delta t + o(\Delta t)$$

④ The probability that there is more than one birth or death within the time interval Δt is negligible (not required).

For the model of the birth–death process we have

$$P_n(t + \Delta t) = \Pr\{N(t + \Delta t) = n\}$$
$$= \Pr\{N(t) = n - 1 \text{ and } N(\Delta t) = 1 \quad \text{or} \quad N(t) = n + 1 \text{ and } N(\Delta t) = -1 \quad \text{or} \quad N(t) = n \text{ and } N(\Delta t) = 0\}$$
$$\Rightarrow \quad \Pr\{N(t) = n - 1\} \cdot \Pr\{N(\Delta t) = 1 \mid N(t) = n - 1\}$$
$$+\Pr\{N(t) = n + 1\} \cdot \Pr\{N(\Delta t) = -1 \mid N(t) = n + 1\}$$
$$+\Pr\{N(t) = n\} \cdot \Pr\{N(\Delta t) = 0 \mid N(t) = n\}$$

Substituting the assumptions above (the conditional probabilities),

$$\Rightarrow \quad P_n(t + \Delta t) = P_{n-1}(t)\left[\lambda_{n-1}\Delta t + o(\Delta t)\right] + P_{n+1}(t)\left[\mu_{n+1}\Delta t + o(\Delta t)\right] + P_n(t)\left[1 - \lambda_n \Delta t - \mu_n \Delta t + o(\Delta t)\right]$$

$\Rightarrow$ opening the brackets:

$$P_n(t + \Delta t) = P_{n-1}(t)\lambda_{n-1}\Delta t + P_{n+1}(t)\mu_{n+1}\Delta t + P_n(t) - P_n(t)\left(\lambda_n \Delta t + \mu_n \Delta t\right) + o(\Delta t), \qquad n \geq 1$$

But

$$P_n'(t) = \lim_{\Delta t \to 0} \frac{P_n(t + \Delta t) - P_n(t)}{\Delta t}$$

— outline notebook, p. 11 —

$$\Leftrightarrow \quad P_n'(t) = \lim_{\Delta t \to 0} \frac{P_{n-1}(t)\lambda_{n-1}\Delta t + P_{n+1}(t)\mu_{n+1}\Delta t + P_n(t) - P_n(t)(\lambda_n \Delta t + \mu_n \Delta t) + o(\Delta t) - P_n(t)}{\Delta t}$$
$$= \lim_{\Delta t \to 0} \left\{ \lambda_{n-1} P_{n-1}(t) + \mu_{n+1} P_{n+1}(t) - P_n(t)(\lambda_n + \mu_n) + \frac{o(\Delta t)}{\Delta t} \right\}$$
$$\Rightarrow \quad P_n'(t) = -(\lambda_n + \mu_n)P_n(t) + \lambda_{n-1}P_{n-1}(t) + \mu_{n+1}P_{n+1}(t), \qquad n \geq 1 \qquad \cdots ①$$

and

$$P_0'(t) = -(\lambda_0 + \mu_0)P_0(t) + \mu_1 P_1(t), \qquad n = 0 \qquad \cdots ②$$

- These equations ① and ② are the difference differential equations of a birth–death process. The problem is to solve these equations for the special cases of λ_n and μ_n, subject to some initial conditions.

9.3 The pure birth process (no deaths)

From the difference differential equations of the birth–death process we can get the difference differential equations of a pure birth process, where the death rate is 0:

$$P_n'(t) = -\lambda_n P_n(t) + \lambda_{n-1}P_{n-1}(t), \qquad n \geq 1$$

and

$$P_0'(t) = -\lambda_0 P_0(t), \qquad n = 0$$

$\rightarrow$ These equations can be solved using the probability generating function technique.

Special cases

① For $\lambda_n = \lambda$ we have a **Poisson process**, i.e.

$$P_n'(t) = -\lambda P_n(t) + \lambda P_{n-1}(t), \quad n \geq 1 \qquad \text{and} \qquad P_0'(t) = -\lambda P_0(t), \quad n = 0.$$

② For $\lambda_n = n\lambda$ we have a **simple birth process** or **Yule's process** (Yule–Furry process).

③ For $\lambda_n = \lambda\left(1 + \frac{q_n}{1+\lambda at}\right)$ we have the **Polya process**.

9.4 Simple birth process (Yule process / Yule–Furry process)

— outline notebook, p. 11 (dated Wednesday 12th March 2025) —

In this case the birth rate is $\lambda_n = n\lambda$, so that $\lambda_{n-1} = (n-1)\lambda$, and the death rate $\mu_n = 0$. From the previous differential equations for the pure birth process, the difference differential equations for the simple birth process become

$$P_n'(t) = -n\lambda P_n(t) + (n-1)\lambda P_{n-1}(t), \qquad n \geq 1$$
$$P_0'(t) = 0, \qquad n = 0 \qquad \cdots ①$$

Next is to get the solution: define the p.g.f.

$$G(s,t) = \sum_{n=0}^{\infty} P_n(t)s^n$$

so that

$$\frac{\partial}{\partial t}G(s,t) = \sum_{n=0}^{\infty} P_n'(t)s^n \qquad \text{and} \qquad \frac{\partial}{\partial s}G(s,t) = \sum_{n=0}^{\infty} nP_n(t)s^{n-1}$$

From the first set of difference differential equations we have

$$\sum_{n=1}^{\infty} P_n'(t)s^n = -\lambda \sum_{n=1}^{\infty} nP_n(t)s^n + \lambda \sum_{n=1}^{\infty} (n-1)P_{n-1}(t)s^n$$
$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) - P_0'(t) = -\lambda s\frac{\partial}{\partial s}G(s,t) + \lambda s^2\frac{\partial}{\partial s}G(s,t)$$

But $P_0'(t) = 0$, so

$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) = -\lambda s(1-s)\frac{\partial}{\partial s}G(s,t)$$
$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) + \lambda s(1-s)\frac{\partial}{\partial s}G(s,t) = 0$$

is the Lagrange's linear equation, and the corresponding auxiliary system of equations is

— outline notebook, p. 12 —

$$\frac{\partial t}{1} = \frac{\partial s}{\lambda s(1-s)} = \frac{dG(s,t)}{0}$$

Equating the first two equations:

$$\frac{\partial t}{1} = \frac{\partial s}{\lambda s(1-s)}$$
$$\Rightarrow \quad \lambda\, dt = \frac{ds}{s(1-s)}$$
$$\Rightarrow \quad \int \lambda\, dt = \int \frac{ds}{s} + \int \frac{ds}{1-s}$$
$$\Rightarrow \quad \lambda t + c_1 = \ln s - \ln(1-s) = \ln\left(\frac{s}{1-s}\right)$$
$$\Rightarrow \quad e^{\lambda t + c_1} = \frac{s}{1-s} \quad \Rightarrow \quad c_1 = e^{\lambda t}\frac{s}{1-s}$$

Let

$$u = e^{\lambda t}\frac{s}{1-s};$$

and from the second equation $\frac{dG(s,t)}{0} \Rightarrow G(s,t) = c_2$.

The general solution is

$$\Phi(c_1, c_2) = 0, \qquad \text{or better,} \qquad c_2 = \Phi(c_1) = \Phi\left(\frac{se^{\lambda t}}{1-s}\right)$$
$$\Rightarrow \quad G(s,t) = \Phi\left(\frac{se^{\lambda t}}{1-s}\right)$$

Using the boundary condition:

$$G(s,0) = \Phi\left(\frac{s}{1-s}\right)$$

But from the definition of the p.g.f.,

$$G(s,t) = \sum_{n=0}^{\infty} P_n(t)s^n \quad \Rightarrow \quad G(s,0) = \sum_{n=0}^{\infty} P_n(0)s^n = P_0(0) + P_1(0)s + \cdots$$

For a simple birth process the initial conditions are

$$P_n(0) = 1 \text{ for } n = 1 \qquad \text{and} \qquad P_n(0) = 0 \text{ for } n \neq 1,$$
$$\Rightarrow \quad G(s,0) = s$$

so

$$\Phi\left(\frac{se^{\lambda t}}{1-s}\right) = s$$

Let us use the change of variable: let

$$x = \frac{se^{\lambda t}}{1-s} \quad \Rightarrow \quad x(1-s) = se^{\lambda t} \quad \Rightarrow \quad x = se^{\lambda t} + sx = s\left(e^{\lambda t} + x\right)$$
$$\Rightarrow \quad s = \frac{x}{e^{\lambda t} + x}$$
$$\Rightarrow \quad \Phi(x) = \frac{e^{-\lambda t}x}{1 + xe^{-\lambda t}}$$
$$\Rightarrow \quad G(s,t) = \Phi\left(\frac{se^{\lambda t}}{1-s}\right) = \frac{e^{-\lambda t} \cdot \frac{se^{\lambda t}}{1-s}}{1 + \frac{se^{-\lambda t}}{1-s}} = \frac{s}{1 - s + se^{-\lambda t}}$$
$$\therefore \quad G(s,t) = \frac{e^{-\lambda t}s}{1 - \left(1 - e^{-\lambda t}\right)s} \qquad \cdots ⑥$$

Expanding,

$$G(s,t) = e^{-\lambda t}s\left[1 - \left(1 - e^{-\lambda t}\right)s\right]^{-1}$$
$$= e^{-\lambda t}s\left[1 + \left(1 - e^{-\lambda t}\right)s + \left(1 - e^{-\lambda t}\right)^2 s^2 + \cdots + \left(1 - e^{-\lambda t}\right)^{n-1} s^{n-1} + \left(1 - e^{-\lambda t}\right)^n s^n + \cdots\right]$$

Therefore the solution of the process is $P_n(t)$, the coefficient of s^n in the expansion of the p.g.f. $G(s,t)$:

$$P_n(t) = e^{-\lambda t}\left(1 - e^{-\lambda t}\right)^{n-1}, \qquad n \geq 1$$

— outline notebook, pp. 16–17 —

9.4a A second pass at the pure birth solution (notebook pp. 16–17)

The outline notebook re-works the pure birth solution a second time, writing ξ for λ and taking the initial state to be 1 instead of 0. It is reproduced here because it records a different route to the same p.g.f.

Pure birth process. For $n \geq 1$:

$$P_n'(t) = -\xi P_n(t) + \xi P_{n-1}(t), \qquad P_0'(t) = -\xi P_0(t), \quad n = 0$$

With

$$G(s,t) = \sum_{n=0}^{\infty} P_n(t)s^n, \qquad \frac{\partial}{\partial t}G(s,t) = \sum_{n=0}^{\infty} P_n'(t)s^n, \qquad \frac{\partial}{\partial s}G(s,t) = \sum_{n=0}^{\infty} nP_n(t)s^{n-1}$$
$$\sum_{k=1}^{\infty} P_n'(t)s^n = -\xi \sum P_n(t)s^n + \xi s \sum P_{n-1}(t)s^{n-1}$$
$$G(s,t) - P_0'(t) = -\xi\left(G(s,t) - P_0(t)\right) + s\xi G(s,t)$$
$$G(s,t) = -\xi G(s,t) + s\xi G(s,t) = -\xi G(s,t)(1-s)$$

Since $P_0'(t) = 0$,

$$1 \cdot \frac{\partial G(s,t)}{\partial t} + 0 \cdot \frac{\partial G(s,t)}{\partial s} = -\xi G(s,t)(1-s)$$
$$\frac{dt}{1} = \frac{ds}{0} = \frac{dG(s,t)}{-\xi G(s,t)(1-s)}$$
$$\int dt = \int \frac{dG(s,t)}{-\xi G(s,t)(1-s)} \quad \Rightarrow \quad -\xi(1-s)t + c = \ln G(s,t)$$

$$G(s,t) = e^{-\xi(1-s)t + c}$$

$$G(s,0) = \sum_{n=0}^{\infty} P_n(0)s^n = P_0(0) + P_1(0)s + P_2(0)s^2 + \cdots = s$$

$$e^c = s \quad \Rightarrow \quad c = \ln s$$

so

$$G(s,t) = e^{-\xi(1-s)t + \ln s} = e^{-\xi(1-s)t} \cdot e^{\ln s} = s\, e^{-\xi(1-s)t} = s\, e^{-\xi t + \xi s t}$$

which can also be written as

$$s\, e^{\xi t}\left[1 + \xi t + \frac{(\xi t)^2}{2!} + \cdots\right] = \frac{s\, e^{\xi t}(\xi t)^n}{n!} = \frac{e^{-\xi t}(\xi t)^n}{n!}$$

[The notebook leaves the last line in this form; reading off the coefficient of s^n in $G(s,t) = se^{-\xi(1-s)t}$ gives $P_n(t) = \frac{e^{-\xi t}(\xi t)^{n-1}}{(n-1)!}$ for $n \geq 1$, the one-term-shifted Poisson law.]

9.5 The linear birth–death process

— outline notebook, p. 13 —

Two cases shall be considered, i.e.

① The simple birth–death process ② The zero growth process

9.5.1 The simple birth–death process

In this case the birth rate is $\lambda_n = n\lambda$ and the death rate is $\mu_n = n\mu$. So the difference differential equations become

$$P_n'(t) = -n(\lambda+\mu)P_n(t) + (n-1)\lambda P_{n-1}(t) + (n+1)\mu P_{n+1}(t), \qquad n \geq 1$$
$$P_0'(t) = \mu P_1(t), \qquad n = 0 \qquad \cdots ①$$

Next we get the solution. Determine the p.g.f.

$$G(s,t) = \sum_{n=0}^{\infty} P_n(t)s^n$$

so that

$$\frac{\partial}{\partial t}G(s,t) = \sum_{n=0}^{\infty} P_n'(t)s^n \qquad \text{and} \qquad \frac{\partial}{\partial s}G(s,t) = \sum_{n=0}^{\infty} nP_n(t)s^{n-1}$$

$$\Rightarrow \quad \sum_{n=1}^{\infty} P_n'(t)s^n = -(\lambda+\mu)\sum_{n=1}^{\infty} nP_n(t)s^n + \lambda\sum_{n=1}^{\infty}(n-1)P_{n-1}(t)s^n + \mu\sum_{n=1}^{\infty}(n+1)P_{n+1}(t)s^n$$

$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) - P_0'(t) = -(\lambda+\mu)s\sum_{n=1}^{\infty} nP_n(t)s^{n-1} + \lambda s^2\sum_{n=1}^{\infty}(n-1)P_{n-1}(t)s^{n-2} + \mu\left[\frac{\partial}{\partial s}G(s,t) - P_1(t)\right]$$

$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) - P_0'(t) = -(\lambda+\mu)s\frac{\partial}{\partial s}G(s,t) + \lambda s^2\frac{\partial}{\partial s}G(s,t) + \mu\frac{\partial}{\partial s}G(s,t) - \mu P_1(t)$$

But from the second set of the difference differential equations (①),

$$P_0'(t) = \mu P_1(t)$$

$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) = \left[-(\lambda+\mu)s + \lambda s^2 + \mu\right]\frac{\partial}{\partial s}G(s,t)$$

$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) = (1-s)(\mu - \lambda s)\frac{\partial}{\partial s}G(s,t)$$

$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) - (1-s)(\mu-\lambda s)\frac{\partial}{\partial s}G(s,t) = 0$$

is the Lagrange's linear equation; the corresponding auxiliary set of equations are

— outline notebook, p. 13 (right column) —

$$\frac{dt}{1} = \frac{ds}{-(1-s)(\mu-\lambda s)} = \frac{dG(s,t)}{0}$$

Consider

$$\frac{dt}{1} = \frac{dG(s,t)}{0} \quad \Rightarrow \quad \int dt = \int \frac{dG(s,t)}{0} \quad \Rightarrow \quad G(s,t) = c_1$$

Next consider

$$\frac{dt}{1} = \frac{ds}{-(1-s)(\mu-\lambda s)} \quad \Rightarrow \quad \int dt = \int \frac{ds}{(1-s)(\mu-\lambda s)}$$

and to integrate we use partial fractions:

$$\frac{1}{(1-s)(\mu-\lambda s)} = \frac{1}{\lambda-\mu}\left[\frac{-\lambda}{\lambda s-\mu}+\frac{1}{1-s}\right]$$

$$\Rightarrow \quad \int dt = \frac{1}{\lambda-\mu}\left[\int \frac{-\lambda\, ds}{\lambda s-\mu}+\int \frac{ds}{1-s}\right]$$

$$\Rightarrow \quad (t+c_3)(\lambda-\mu) = -\ln(\lambda s-\mu)-\ln(1-s)$$

$$\Rightarrow \quad \ln\left(\frac{\lambda s-\mu}{1-s}\right) = -(\lambda-\mu)t-(\lambda-\mu)c$$

$$\Rightarrow \quad \frac{\lambda s-\mu}{1-s} = e^{-(\lambda-\mu)t}\cdot e^{-(\lambda-\mu)c}$$

$$\Rightarrow \quad e^{(\lambda-\mu)c} = c_2 = \frac{\lambda s-\mu}{1-s}\cdot e^{(\lambda-\mu)t}$$

So the general solution of the Lagrange's linear equation is

$$c_1 = \Psi(c_2)$$

$$\Rightarrow \quad G(s,t) = \Psi\left(\frac{\lambda s-\mu}{1-s}e^{-(\lambda-\mu)t}\right) \qquad \cdots ②$$

If the initial conditions are

$$P_n(0) = 1, \quad n = 1 \qquad \text{and} \qquad P_n(0) = 0, \quad n \neq 1,$$

then from ②

$$G(s,0) = \Psi\left(\frac{\lambda s-\mu}{1-s}\right) \qquad \cdots ③$$

Again

$$G(s,t) = P_0(t)+P_1(t)s+P_2(t)s^2+P_3(t)s^3+\cdots$$

$$\Rightarrow \quad G(s,0) = P_0(0)+P_1(0)s+P_2(0)s^2+\cdots = P_1(0)s = s$$

$$\Rightarrow \quad G(s,0) = \Psi\left(\frac{\lambda s-\mu}{1-s}\right) = s$$

— outline notebook, p. 14 —

We let

$$\omega = \frac{\lambda s-\mu}{1-s} \quad \Rightarrow \quad s = \frac{\mu+\omega}{\lambda+\omega}$$

$$\Rightarrow \quad \text{If } \Psi\left(\frac{\lambda s-\mu}{1-s}\right) = s \qquad \text{then} \qquad \Psi(\omega) = \frac{\mu+\omega}{\lambda+\omega}$$

From equation ②, our $G(s,t)$ is

$$G(s,t) = \Psi\left(\omega e^{-(\lambda-\mu)t}\right) = \frac{\mu+\omega e^{-(\lambda-\mu)t}}{\lambda+\omega e^{-(\lambda-\mu)t}}$$

$$= \frac{\mu+\frac{\lambda s-\mu}{1-s}e^{-(\lambda-\mu)t}}{\lambda+\frac{\lambda s-\mu}{1-s}e^{-(\lambda-\mu)t}} \qquad \times(1-s)$$

$$= \frac{(1-s)\mu e^{-(\lambda-\mu)t}+\lambda s-\mu}{(1-s)\lambda e^{-(\lambda-\mu)t}+\lambda s-\mu}$$

$$= \frac{(s-1)\mu e^{-(\lambda-\mu)t}+\mu-\lambda s}{(s-1)\lambda e^{-(\lambda-\mu)t}+\lambda s-\mu}$$

$$= \frac{\mu\left[1-e^{-(\lambda-\mu)t}\right]-s\left[\lambda-\mu e^{-(\lambda-\mu)t}\right]}{\mu-\lambda e^{-(\lambda-\mu)t}-\lambda s\left[1-e^{-(\lambda-\mu)t}\right]}$$

Writing

$$\phi(t) = \frac{1-e^{-(\lambda-\mu)t}}{\mu-\lambda e^{-(\lambda-\mu)t}}$$

and dividing numerator and denominator by $\mu-\lambda e^{-(\lambda-\mu)t}$:

$$G(s,t) = \frac{\mu\phi(t)+s\left[1-(\lambda+\mu)\phi(t)\right]}{1-\lambda s\,\phi(t)} \qquad \cdots ④$$

where the ratio

$$\frac{\lambda-\mu e^{-(\lambda-\mu)t}}{\mu-\lambda e^{-(\lambda-\mu)t}} = -1+(\lambda+\mu)\phi(t)$$

was used for the coefficient of s (obtained by taking the LCM of the two fractions and simplifying).

Therefore, back to equation ④,

$$G(s,t) = \frac{\mu\phi(t) + s\big(1 - (\lambda+\mu)\phi(t)\big)}{1 - \lambda s\phi(t)} = \big[\mu\phi(t) + s\big(1 - (\lambda+\mu)\phi(t)\big)\big] \sum_{n=0}^{\infty} \big(\lambda s\phi(t)\big)^n$$

So the solution of the process, i.e. $P_n(t)$, is the coefficient of s^n in the $G(s,t)$ expansion:

$$P_n(t) = \mu\phi(t) \cdot \big[\lambda\phi(t)\big]^n + \big[1 - (\lambda+\mu)\phi(t)\big]\big[\lambda\phi(t)\big]^{n-1}$$

which can be left at this, or simplified to

$$= \big[\lambda\phi(t)\big]^{n-1}\big[\big(1 - \lambda\phi(t)\big)\big(1 - \mu\phi(t)\big)\big]$$

[The outline notebook heads this page "Feller's method" and continues with that method on p. 18.]

9.6 The zero growth process

— outline notebook, p. 20 —

Here the birth rate is $\lambda_n = n\lambda$, the death rate is $\mu_n = n\mu$ and $\lambda = \mu$. So the difference differential equations for this process are

$$P_n'(t) = -2n\lambda P_n(t) + (n-1)\lambda P_{n-1}(t) + (n+1)\lambda P_{n+1}(t), \qquad n \geq 1$$
$$P_0'(t) = \lambda P_1(t), \qquad n = 0$$

To get the solution, as usual define the p.g.f.

$$G(s,t) = \sum_{n=0}^{\infty} P_n(t)s^n$$

$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) = \sum_{n=0}^{\infty} P_n'(t)s^n \qquad \text{and} \qquad \frac{\partial}{\partial s}G(s,t) = \sum_{n=0}^{\infty} nP_n(t)s^{n-1}$$

$$\Rightarrow \quad \sum_{n=1}^{\infty} P_n'(t)s^n = -2\lambda \sum_{n=1}^{\infty} nP_n(t)s^n + \lambda \sum_{n=1}^{\infty} (n-1)P_{n-1}(t)s^n + \lambda \sum_{n=1}^{\infty} (n+1)P_{n+1}(t)s^n$$

$$\frac{\partial}{\partial t}G(s,t) - P_0'(t) = -2\lambda s \sum_{n=1}^{\infty} nP_n(t)s^{n-1} + \lambda s^2 \sum_{n=1}^{\infty} (n-1)P_{n-1}(t)s^{n-2} + \lambda\left[\frac{\partial}{\partial s}G(s,t) - P_1(t)\right]$$

Since $P_0'(t) = \lambda P_1(t)$ we have

$$\frac{\partial}{\partial t}G(s,t) = -2\lambda s\frac{\partial}{\partial s}G(s,t) + \lambda s^2\frac{\partial}{\partial s}G(s,t) + \lambda\frac{\partial}{\partial s}G(s,t)$$
$$= \big(\lambda - 2\lambda s + \lambda s^2\big)\frac{\partial}{\partial s}G(s,t) = \lambda(1-s)^2\frac{\partial}{\partial s}G(s,t)$$
$$\Rightarrow \quad \frac{\partial}{\partial t}G(s,t) - \lambda(1-s)^2\frac{\partial}{\partial s}G(s,t) = 0$$

is the Lagrange's linear equation, and the corresponding auxiliary set of equations are

$$\frac{dt}{1} = \frac{ds}{-\lambda(1-s)^2} = \frac{dG(s,t)}{0}$$

Consider $\frac{dt}{1} = \frac{ds}{-\lambda(1-s)^2}$:

$$\Rightarrow \quad \int \frac{dt}{1} = \int \frac{ds}{-\lambda(1-s)^2} \quad \Leftrightarrow \quad \int -\lambda\, dt = \int \frac{ds}{(1-s)^2}$$
$$\Rightarrow \quad -\lambda t = \frac{1}{1-s} \quad \Rightarrow \quad -\lambda t + c_1 = \frac{1}{1-s}$$

making c the subject,

$$\Rightarrow \quad c_1 = \frac{1}{1-s} + \lambda t$$

Next consider $\frac{dt}{1} = \frac{dG(s,t)}{0}$:

$$\Rightarrow \quad \int dt = \int \frac{dG(s,t)}{0} \quad \Rightarrow \quad \int 0\, dt = \int dG(s,t) \quad \Rightarrow \quad G(s,t) = c_2$$

So the general solution of the Lagrange's linear equation is

$$c_2 = \Psi(c_1) \quad \Rightarrow \quad G(s,t) = \Psi\left(\frac{1}{1-s} + \lambda t\right) \qquad \cdots ⑤$$

Suppose the initial conditions are

$$P_n(0) = 1, \quad n = 1 \qquad \text{or} \qquad 0, \quad n \neq 1,$$

then

$$G(s,t) = P_0(t) + P_1(t)s + P_2(t)s^2 + \cdots$$
$$G(s,0) = P_0(0) + P_1(0)s + P_2(0)s^2 + \cdots = P_1(0)s = s$$

From equation ⑤,

$$\Rightarrow \quad G(s,0) = \Psi\left(\frac{1}{1-s}\right) = s$$

Suppose we introduce a transformation by letting

$$\omega = \frac{1}{1-s} \quad \Rightarrow \quad s = \frac{\omega - 1}{\omega} \quad \Rightarrow \quad \Psi(\omega) = \frac{\omega - 1}{\omega}$$

From equation ⑤,

$$G(s,t) = \Psi(\omega + \lambda t) = \frac{\omega + \lambda t - 1}{\omega + \lambda t} = \frac{\frac{1}{1-s} + \lambda t - 1}{\frac{1}{1-s} + \lambda t} \qquad \times (1-s)$$

$$\Rightarrow \quad G(s,t) = \frac{1 + \lambda t(1-s) - (1-s)}{1 + \lambda t(1-s)} = \frac{\lambda t - \lambda t s + s}{1 + \lambda t - \lambda t s}$$

$$\therefore \quad G(s,t) = \frac{\lambda t + (1 - \lambda t)s}{1 + \lambda t - \lambda t s} \qquad \cdots ⑥$$

— outline notebook, p. 21 —

$$G(s,t) = \frac{\lambda t + (1-\lambda t)s}{1 + \lambda t} \cdot \left(\frac{1}{1 - \frac{\lambda s}{1 + \lambda t}}\right) \qquad \text{provided } \left|\frac{\lambda s}{1 + \lambda t}\right| < 1$$

- The solution of the process, denoted $P_n(t)$, is actually the coefficient of s^n in the expansion of $G(s,t)$. But

$$G(s,t) = \frac{\lambda t + (1-\lambda t)s}{1 + \lambda t}\left\{1 + \left(\frac{\lambda s}{1 + \lambda t}\right) + \left(\frac{\lambda s}{1 + \lambda t}\right)^2 + \left(\frac{\lambda s}{1 + \lambda t}\right)^3 + \cdots + \left(\frac{\lambda s}{1 + \lambda t}\right)^{n-1} + \left(\frac{\lambda s}{1 + \lambda t}\right)^n + \cdots\right\}$$

⇒ Thus the solution $P_n(t)$:

$$P_n(t) = \left(\frac{\lambda t}{1 + \lambda t}\right)\left(\frac{\lambda t}{1 + \lambda t}\right)^n + \left(\frac{1 - \lambda t}{1 + \lambda t}\right)\left(\frac{\lambda t}{1 + \lambda t}\right)^{n-1}$$

$$= \frac{(\lambda t)^{n+1}}{(1 + \lambda t)^{n+1}} + \frac{(\lambda t)^{n-1} - (\lambda t)^n}{(1 + \lambda t)^n} = \frac{(\lambda t)^{n+1} + (\lambda t)^{n-1}(1 + \lambda t) - (\lambda t)^n(1 + \lambda t)}{(1 + \lambda t)^{n+1}}$$

$$\Rightarrow \quad P_n(t) = \frac{(\lambda t)^{n+1} + (\lambda t)^{n-1} + (\lambda t)^n - (\lambda t)^n - (\lambda t)^{n+1}}{(1 + \lambda t)^{n+1}} = \frac{(\lambda t)^{n-1}}{(1 + \lambda t)^{n+1}}, \qquad n \geq 1$$

The mean of the process:

$$E(N) = G'(1,t) = 1 \quad \rightarrow \text{show}$$

The variance of the process:

$$\text{Var}(N) = G''(1,t) + G'(1,t) - \left(G'(1,t)\right)^2 = 2\lambda t \quad \rightarrow \text{show}$$

Hint: use equation ⑥ and the quotient/product rule.

9.7 Feller's method: mean and variance of the simple birth–death process

— outline notebook, p. 18 (dated Wednesday 26th March 2025) —

Defining the functions:

$$M_1(t) = \sum_{n=0}^{\infty} nP_n(t), \qquad M_2(t) = \sum_{n=0}^{\infty} n^2 P_n(t), \qquad M_3(t) = \sum_{n=0}^{\infty} n^3 P_n(t), \ldots$$

so that

$$M_1'(t) = \sum_{n=0}^{\infty} nP_n'(t), \qquad M_2'(t) = \sum_{n=0}^{\infty} n^2 P_n'(t), \qquad M_3'(t) = \sum_{n=0}^{\infty} n^3 P_n'(t)$$

From the first set of the difference differential equations of the simple birth–death process, i.e.

$$P_n'(t) = -n(\lambda + \mu)P_n(t) + (n-1)\lambda P_{n-1}(t) + (n+1)\mu P_{n+1}(t), \qquad n \geq 1 \qquad \cdots ①$$

from this equation ① we introduce n and sum over n to get

$$\sum_{n=1}^{\infty} nP_n'(t) = -(\lambda+\mu)\sum_{n=1}^{\infty} n^2P_n(t) + \lambda\sum_{n=1}^{\infty} n(n-1)P_{n-1}(t) + \mu\sum_{n=1}^{\infty} n(n+1)P_{n+1}(t)$$

$$M_1'(t) = -(\lambda+\mu)M_2(t) + \lambda\sum_{n=1}^{\infty}\left\{(n-1)^2 + (n-1)\right\}P_{n-1}(t) + \mu\sum_{n=1}^{\infty}\left\{(n+1)^2 - (n+1)\right\}P_{n+1}(t)$$

$$\Rightarrow \quad M_1'(t) = -\lambda M_2(t) - \mu M_2(t) + \lambda\sum(n-1)^2P_{n-1}(t) + \lambda\sum(n-1)P_{n-1}(t) + \mu\sum(n+1)^2P_{n+1}(t) - \mu\sum(n+1)P_{n+1}(t)$$

$$\Rightarrow \quad M_1'(t) = -\lambda M_2(t) - \mu M_2(t) + \lambda M_2(t) + \lambda M_1(t) + \mu\{M_2(t) - P_1(t)\} - \mu\{M_1(t) - P_0(t)\}$$

$$\Rightarrow \quad M_1'(t) = (\lambda-\mu)M_1(t)$$

$$\Rightarrow \quad \frac{M_1'(t)}{M_1(t)} = \int(\lambda-\mu)\,dt$$

$$= \ln M_1(t) = (\lambda-\mu)t + c$$

— outline notebook, p. 18 (right column) —

$$M_1(t) = e^{(\lambda-\mu)t+c} \qquad \cdots ②$$

If the initial conditions are

$$P_n(0) = 1, \quad n = 1 \qquad \text{and} \qquad P_n(0) = 0, \quad n \neq 1,$$

then

$$M_1(t) = \sum_{n=0}^{\infty} nP_n(t) = 0 + P_1(t) + 2P_2(t) + 3P_3(t) + \cdots$$

$$\Rightarrow \quad M_1(0) = P_1(0) + 2P_2(0) + 3P_3(0) + \cdots = P_1(0) = 1$$

Again from equation ②,

$$M_1(0) = e^c = 1 \quad \Rightarrow \quad c = 0$$

Therefore

$$M_1(t) = e^{(\lambda-\mu)t}$$

which is actually the mean of the process, since

$$E(N) = \sum_{n=0}^{\infty} nP_n(t) = M_1(t)$$

Next we get the variance of the process. From equation ① we introduce both n and n^2 and sum over n to get

$$\sum_{n=1}^{\infty} n^2P_n'(t) = -(\lambda+\mu)\sum_{n=1}^{\infty} n^3P_n(t) + \lambda\sum_{n=1}^{\infty} n^2(n-1)P_{n-1}(t) + \mu\sum_{n=1}^{\infty} n^2(n+1)P_{n+1}(t)$$

(using $(n-1)^3 = (n-1)^3$ and $(n+1)^3 = (n+1)^3 - 2(n+1)^2 + (n+1)$ as substituted in the notes)

$$\Rightarrow \quad M_2'(t) = -\lambda M_3(t) - \mu M_3(t) + \lambda\sum\left\{(n-1)^3 + 2(n-1)^2 + (n-1)\right\}P_{n-1}(t) + \mu\sum\left\{(n+1)^3 + 2(n+1)^2 + (n+1)\right\}P_{n+1}(t)$$

$$\Rightarrow \quad M_2'(t) = -\lambda M_3(t) - \mu M_3(t) + \lambda\sum(n-1)^3P_{n-1}(t) + 2\lambda\sum(n-1)^2P_{n-1}(t) + \lambda\sum(n-1)P_{n-1}(t)$$

$$+\mu\sum(n+1)^3P_{n+1}(t) - 2\mu\sum(n+1)^2P_{n+1}(t) + \mu\sum(n+1)P_{n+1}(t)$$

— outline notebook, p. 19 —

$$\Rightarrow \quad M_2'(t) = -\lambda M_3(t) - \mu M_3(t) + \lambda M_3(t) + 2\lambda M_2(t) + \lambda M_1(t) + \mu\{M_3(t) - P_1(t)\} - 2\mu\{M_2(t) - P_1(t)\} + \mu\{M_1(t) - P_0(t)\}$$

$$\Rightarrow \quad M_2'(t) = 2(\lambda-\mu)M_2(t) + (\lambda+\mu)M_1(t)$$

$$\Rightarrow \quad M_2'(t) - 2(\lambda-\mu)M_2(t) = (\lambda+\mu)M_1(t) \qquad \cdots ③$$

Note that equation ③ is of the form

$$\frac{dy}{dx} + P(x)y = Q(x),$$

a linear equation which can be solved using the integrating factor (I.F.).

The I.F. in this case is

$$\text{I.F.} = \mu(x) = e^{\int P(x)dx},$$

which is multiplied throughout in the above linear equation.

Back to equation ③, we solve it using the integrating factor:

$$\text{I.F.} = q(t) = e^{-\int 2(\lambda-\mu)dt} = e^{-2(\lambda-\mu)t}$$

We multiply equation ③ by I.F. $= e^{-2(\lambda-\mu)t}$:

$$M_2'(t)\cdot e^{-2(\lambda-\mu)t} - 2(\lambda-\mu)e^{-2(\lambda-\mu)t}\cdot M_2(t) = (\lambda+\mu)e^{-2(\lambda-\mu)t}\cdot e^{(\lambda-\mu)t}$$

since $M_1(t) = e^{(\lambda-\mu)t}$, the mean of the process.

$$\Rightarrow \quad M_2'(t)\cdot e^{-2(\lambda-\mu)t} - 2(\lambda-\mu)e^{-2(\lambda-\mu)t}\cdot M_2(t) = (\lambda+\mu)e^{-(\lambda-\mu)t}$$

Equivalent to (using the product rule)

$$\Leftrightarrow \quad \frac{d}{dt}\left[M_2(t)e^{-2(\lambda-\mu)t}\right] = (\lambda+\mu)e^{-(\lambda-\mu)t}$$

Next is to integrate both sides, to get

$$M_2(t)e^{-2(\lambda-\mu)t} = \int (\lambda+\mu)e^{-(\lambda-\mu)t}dt + c = \frac{(\lambda+\mu)}{-(\lambda-\mu)}e^{-(\lambda-\mu)t} + c \qquad \cdots ④$$

Suppose the initial conditions are

$$P_n(0) = \begin{cases} 1, & n = 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$M_2(t) = \sum_{n=0}^{\infty} n^2 P_n(t) = 0 + P_1(t) + 2^2P_2(t) + 3^2P_3(t) + \cdots$$

$$\Rightarrow \quad M_2(0) = 0 + P_1(0) + 2^2P_2(0) + 3^2P_3(0) + \cdots = P_1(0) = 1$$

From equation ④, let $t = 0$:

$$M_2(0) = \frac{\lambda+\mu}{-(\lambda-\mu)} + c = 1, \qquad \text{making } c \text{ the subject} \quad \Rightarrow \quad c = 1 + \frac{\lambda+\mu}{\lambda-\mu}$$

Therefore, from the same equation ④,

$$M_2(t) = \frac{\lambda+\mu}{-(\lambda-\mu)}e^{(\lambda-\mu)t} + \left(1 + \frac{\lambda+\mu}{\lambda-\mu}\right)e^{2(\lambda-\mu)t}$$

Therefore the variance of the process is

$$\text{Var}(N) = E(N^2) - \big(E(N)\big)^2 = \sum_{n=0}^{\infty} n^2P_n(t) - \left(\sum_{n=0}^{\infty} nP_n(t)\right)^2 = M_2(t) - \big(M_1(t)\big)^2$$

$$= -\frac{\lambda+\mu}{\lambda-\mu}e^{(\lambda-\mu)t} + \left(1 + \frac{\lambda+\mu}{\lambda-\mu}\right)e^{2(\lambda-\mu)t} - \left(e^{(\lambda-\mu)t}\right)^2$$

$$= -\frac{\lambda+\mu}{\lambda-\mu}e^{(\lambda-\mu)t} + \frac{\lambda+\mu}{\lambda-\mu}e^{2(\lambda-\mu)t}$$

$$\therefore \quad \text{Var}(N) = \frac{\lambda+\mu}{\lambda-\mu}e^{(\lambda-\mu)t}\left[e^{(\lambda-\mu)t} - 1\right]$$

— outline notebook, p. 21 (dated Wednesday 2nd April 2025) —

10. Queuing Models

- A queue (waiting line) is formed when customers arrive at a service facility and wait for a service.
- A queue is formed when there are more customers than the server can handle. In a queuing process a customer arrives and queues (to wait) if all the servers are fully occupied, eventually receives service and finally departs from the facility.
- **Def:** A queuing system is a set of customers, a set of servers and an order whereby customers arrive and get served.
- The following are examples of a queuing system:
  1. People queuing in a cafeteria.
  2. Customers waiting for a service in an insurance office.
  3. People waiting in a doctor's office for consultation.
- In a queuing system, additional service facilities would be provided to decrease the queue or prevent the queue from forming, at a cost, which may result in a decrease in profits to unacceptable levels.
- On the other hand, excessively long queues may result in loss of sales and customers; therefore the rationale of studying queues is to come up with a mathematical model which will enable one to improve the system by balancing

the cost associated with waiting and the cost associated with preventing the queue from building, in order to maximize the profits.

10.1 Characteristics of a queuing system

— outline notebook, p. 22 —

1. Arrival mechanism. This specifies the arrival pattern. Arrival is not generally uniform and the dispersion between the arrival times of the units being served is a random variable with a probability distribution called the inter-arrival time distribution; this distribution is always presumed to be known.

The no. of units arriving in a system per unit time is denoted by λ, and $\frac{1}{\lambda}$ is the average of the inter-arrival time.

2. The queue discipline. Queue discipline is the manner in which customers form a queue. It can be FIFO (First In, First Out) / first come, first served, e.g. in a cafeteria; LIFO (Last In, First Out), e.g. in industrial service. It can also be on a random basis or a priority basis, e.g. in a hospital.

3. The service mechanism. This describes the service operations; it is specified by the service time, i.e. the time required by one server to serve one customer. The service times of successive customers are taken to be independent random variables and their probability distribution function is called the service time distribution. Assuming that all channels are occupied, the no. of customers that can be served per unit time is called the service rate.

4. Service channel. The service mechanism may have a single channel or a no. of channels; in a multi-channel system, channels may be arranged in series, in parallel or in a combination of the two.

5. Maximum capacity of the queue. The maximum queue line may generally be assumed to be infinite. In some queuing systems, a finite no. of customers is allowed. The system capacity is the maximum no. of customers, both in the service and in the queue, that is permitted in the service facility at the same time. Note that a situation whereby a customer arrives at a facility and decides not to join the queue is called **balking**. If a customer who is already in a queue leaves the queue before receiving service, such a situation is referred to as **reneging**. In queuing systems we assume that these two situations don't happen, in order to make a mathematical inference.

10.2 Notations in queuing systems (Kendall's notation)

Kendall introduced a set of notations which have become standard in the literature of queue models. This notation for specifying a queue characteristic is

$$V/W/X/Y/Z,$$

where

- V — denotes the arrival pattern, i.e. the probability distribution of the inter-arrival time;
- W — denotes the service pattern, i.e. the probability distribution of the service time;
- X — no. of available servers;
- Y — denotes the max. no. of customers allowed in the system;
- Z — the queue discipline.

Note that if Y or Z isn't specified, it is taken to be infinite or FIFO respectively.

10.3 Transient and steady states

— outline notebook, p. 23 —

Generally, the behaviour of the queuing system will depend on time. Such a system is said to be in a **transient state**. This occurs at the early stage when the system still depends on the initial conditions. After sufficient elapse of time (t), i.e. as $t \to \infty$, the

behaviour of the system no longer depends on time t and the system is said to be in a **steady state**.

10.4 A queuing system as a stochastic process

- A queuing system is a birth–death process with a population consisting of customers either waiting for a service or currently receiving a service. A birth occurs when a customer arrives at the service facility and a death occurs when a customer departs from the facility.
- Let λ_i denote the no. of arrivals per unit time when the state of the system is a noted state denoted by S_i, and μ_i be the no. of departures per unit time when the system is at state S_i.
- Let Δt be such a small time interval that during this time either zero or 1 arrival, or zero or 1 departure takes place, and not both. Again, 2 or more arrivals or departures are not allowed.
- This situation can be summarized at the time t and $t + \Delta t$ as follows:

Events	State of the system at time t	Arrival	Departure	State of the system at time $t + \Delta t$
1	n	0	0	n
2	n	1	0	$n+1$
3	$n+1$	0	1	n
4	$n+1$	0	1	n

[In the notebook the entries of the table sit half a row below their labels; the fourth row is written as a repeat of the third.]

- Note that at time $t + \Delta t$ there are n customers.
- The 4 events are mutually exclusive and exhaustive.
- Suppose that:

i. Pr(exactly one arrival in time $\Delta t/S_i) = \lambda\Delta t$

ii. Pr(zero arrival in time $\Delta t/S_i) = 1 - \lambda\Delta t$

iii. Pr(one departure in time $\Delta t/S_i) = \mu\Delta t$

iv. Pr(zero departure in time $\Delta t/S_i) = 1 - \mu\Delta t$

Then the probability of having n customers at time $t + \Delta t$ is given by

$$P_n(t+\Delta t) = P_n(t)(1-\lambda\Delta t)(1-\mu\Delta t) + P_n(t)\left(\lambda\Delta t \cdot \mu\Delta t\right) + P_{n-1}(t)(\lambda\Delta t)(1-\mu\Delta t) + P_{n+1}(t)(1-\lambda\Delta t)\mu\Delta t$$
$$= P_n(t)\left(1-(\lambda+\mu)\Delta t\right) + P_{n-1}(t)\lambda\Delta t + P_{n+1}(t)\mu\Delta t, \qquad n \geq 1$$

But

$$P_n'(t) = \frac{d}{dt}P_n(t) = \lim_{\Delta t \to 0}\frac{P_n(t+\Delta t) - P_n(t)}{\Delta t}$$
$$= \lim_{\Delta t \to 0}\frac{P_n(t) - (\lambda+\mu)\Delta t\, P_n(t) + \lambda\Delta t\, P_{n-1}(t) + \mu\Delta t\, P_{n+1}(t) - P_n(t)}{\Delta t}$$
$$P_n'(t) = \lim_{\Delta t \to 0}\left\{-(\lambda+\mu)P_n(t) + \lambda P_{n-1}(t) + \mu P_{n+1}(t)\right\}$$
$$= -(\lambda+\mu)P_n(t) + \lambda P_{n-1}(t) + \mu P_{n+1}(t), \qquad n \geq 1 \qquad \cdots(1)$$

— outline notebook, p. 24 —

For the empty system at time $t + \Delta t$ we have the following table:

Events	State of the system at time t	Arrival	Departure	State of the system at time $t + \Delta t$
1	0	0	0	0
2	1	0	1	0

$$\Rightarrow \quad P_0(t+\Delta t) = P_0(t)(1-\lambda\Delta t)(1) + P_1(t)(1-\lambda\Delta t)(\mu\Delta t)$$

Note that $1 - \mu\Delta t = 1$ since $\mu\Delta t = 0$ as $\mu = 0$, i.e. there's no customer in the queue and the departure rate is zero.

$$\Rightarrow \quad P_0(t+\Delta t) = P_0(t) - \lambda\Delta t\, P_0(t) + \mu\Delta t\, P_1(t), \qquad n = 0$$

$$\Rightarrow \quad P_0^{'}(t) = \frac{d}{dt}P_0(t) = \lim_{\Delta t \to 0}\frac{P_0(t+\Delta t) - P_0(t)}{\Delta t}$$

$$= \lim_{\Delta t \to 0}\frac{P_0(t) - \lambda \Delta t\, P_0(t) + \mu \Delta t\, P_1(t) - P_0(t)}{\Delta t} = \lim_{\Delta t \to 0}\left\{-\lambda P_0(t) + \mu P_1(t)\right\}$$

$$P_0^{'}(t) = \mu P_1(t) - \lambda P_0(t), \qquad n = 0 \qquad \cdots ②$$

Equations ① and ② above are the difference differential equations of a queuing model.

10.5 Steady-state solutions

- Here we get the solutions of the above difference differential equations when the system is at steady state, i.e. when the system is independent of time t.
- The steady-state (equilibrium) probabilities for a queuing system are

$$P_n = \lim_{t \to \infty} P_n(t), \qquad n = 0, 1, 2, \ldots$$

if the limit exists.

- We note that the steady-state probabilities are independent of time t. Since any queuing system operates as a function of time t, we must decide whether we are interested in analysing the system under transient or steady-state conditions. The transient condition prevails when the behaviour of the system continues to depend on time t. Queues with combined arrivals & departures start under transient conditions and gradually reach steady conditions after a sufficient elapse of time t, provided the parameters of the system permit reaching steady state.
- Note that if the arrival rate λ is greater than the departure rate μ, i.e. $\lambda > \mu$, the system will never reach steady state, since the queue size increases with time. Again, if $\lambda < \mu$ the system will always reach steady state.

The steady-state solution can be proved to exist as t approaches ∞ ($t \to \infty$):

$$P_n^{'}(t) \to 0 \qquad \text{and} \qquad P_n(t) \to P_n, \qquad n = 0, 1, 2, \ldots$$

- We can now get the steady-state solutions with equations ① and ② as follows:

$$-(\lambda + \mu)P_n + \mu P_{n+1} + \lambda P_{n-1} = 0 \qquad \cdots ③$$

and

$$\mu P_1 - \lambda P_0 = 0 \qquad \cdots ④$$

- From equation ④, $P_1 = \frac{\lambda}{\mu}P_0$ is the probability of having 1 unit in the system, where λ is the arrival rate into the system, μ is the departure rate from the system and P_0 is the probability that the system is empty.

— outline notebook, p. 25 —

From equation ③ for $n = 1$ we have

$$-(\lambda + \mu)P_1 + \mu P_2 + \lambda P_0 = 0$$

On solving for P_2 we have

$$\mu P_2 = (\lambda + \mu)P_1 - \lambda P_0, \qquad \text{but } P_1 = \frac{\lambda}{\mu}P_0, \text{ so}$$

$$\mu P_2 = (\lambda + \mu)\frac{\lambda}{\mu}P_0 - \lambda P_0$$

$$\mu P_2 = \frac{\lambda^2}{\mu}P_0 = \left(\frac{\lambda}{\mu}\right)^2 P_0 = P_2$$

thus for the general case

$$P_n = \left(\frac{\lambda}{\mu}\right)^n P_0,$$

and the problem is to obtain P_0.

10.6 Determination of P_0

Two cases shall be considered.

Case one: the queue is infinite

Here we assume that the queue is infinite. Now the sum of probabilities, i.e.

$$\sum_{n=0}^{\infty} P_n = 1 \quad \Rightarrow \quad \sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^n P_0 = 1$$

$$\Rightarrow \quad P_0 \sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^n = 1 \quad \Rightarrow \quad P_0\left[1 + \frac{\lambda}{\mu} + \left(\frac{\lambda}{\mu}\right)^2 + \left(\frac{\lambda}{\mu}\right)^3 + \cdots\right] = 1$$

$$\Rightarrow \quad P_0 \cdot \frac{1}{1 - \frac{\lambda}{\mu}} = 1 \quad \Rightarrow \quad P_0 = 1 - \frac{\lambda}{\mu} = 1 - \theta$$

$$\Rightarrow \quad P_n = \left(\frac{\lambda}{\mu}\right)^n P_0 = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right) = \theta^n(1 - \theta)$$

Case two: the queue is finite

When the queue is assumed to be finite, the queue is not allowed to have more than N units, i.e. $n \leq N$. So the sum of probabilities is

$$\sum_{n=0}^{N} P_n = 1 \quad \Rightarrow \quad \sum_{n=0}^{N} \left(\frac{\lambda}{\mu}\right)^n P_0 = 1 \quad \Rightarrow \quad P_0 \sum_{n=0}^{N} \left(\frac{\lambda}{\mu}\right)^n = 1$$

[margin note: $S_\infty = \frac{a(1-r^n)}{1-r}$]

$$\Rightarrow \quad P_0\left[1 + \frac{\lambda}{\mu} + \left(\frac{\lambda}{\mu}\right)^2 + \cdots + \left(\frac{\lambda}{\mu}\right)^N\right] = 1$$

$$\Rightarrow \quad P_0 \cdot \frac{1 - \left(\frac{\lambda}{\mu}\right)^{N+1}}{1 - \frac{\lambda}{\mu}} = 1$$

making P_0 the subject,

$$\Rightarrow \quad P_0 = \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^{N+1}} = \frac{1 - \theta}{1 - \theta^{N+1}}$$

Therefore

$$P_n = \left(\frac{\lambda}{\mu}\right)^n P_0 = \left(\frac{\lambda}{\mu}\right)^n \cdot \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^{N+1}} = \theta^n \cdot \frac{1 - \theta}{1 - \theta^{N+1}}$$

10.7 Parameters associated with an infinite queue

— outline notebook, p. 25 (bottom) —

1. Expected no. of customers in the system (average no. of units in the system)

This is denoted by N_s, where

$$N_s = E(N) = \sum_{n=0}^{\infty} nP_n = \sum_{n=0}^{\infty} n\left(\frac{\lambda}{\mu}\right)^n P_0$$

$$\Rightarrow \quad N_s = P_0 \sum_{n=0}^{\infty} n\left(\frac{\lambda}{\mu}\right)^n = P_0\left[\frac{\lambda}{\mu} + 2\left(\frac{\lambda}{\mu}\right)^2 + 3\left(\frac{\lambda}{\mu}\right)^3 + \cdots\right]$$

$$= P_0 \cdot \frac{\lambda}{\mu}\left[1 + 2\left(\frac{\lambda}{\mu}\right) + 3\left(\frac{\lambda}{\mu}\right)^2 + \cdots\right] = P_0 \cdot \frac{\lambda}{\mu} \cdot f'\left(\frac{\lambda}{\mu}\right)$$

where

— outline notebook, p. 26 —

$$\Rightarrow \quad f'\left(\frac{\lambda}{\mu}\right) = \frac{\lambda}{\mu} + \left(\frac{\lambda}{\mu}\right)^2 + \left(\frac{\lambda}{\mu}\right)^3 + \cdots = \frac{\lambda}{\mu}\left[1 + \frac{\lambda}{\mu} + \left(\frac{\lambda}{\mu}\right)^2 + \cdots\right]$$

use of the quotient rule to differentiate

$$\frac{d}{d\theta}\left(\frac{\lambda}{\mu}\right) = \frac{\lambda}{\mu}\cdot\frac{1}{1-\frac{\lambda}{\mu}} = \frac{\frac{\lambda}{\mu}}{1-\frac{\lambda}{\mu}}$$

$$\frac{d}{d\theta}\left(\frac{\lambda}{\mu}\right) = \frac{1-\frac{\lambda}{\mu}+\frac{\lambda}{\mu}}{\left(1-\frac{\lambda}{\mu}\right)^2} = \frac{1}{\left(1-\frac{\lambda}{\mu}\right)^2}$$

Therefore,

$$N_s = P_0\cdot\left(\frac{\lambda}{\mu}\right)\cdot f'\left(\frac{\lambda}{\mu}\right) = \left(1-\frac{\lambda}{\mu}\right)\left(\frac{\lambda}{\mu}\right)\cdot\frac{1}{\left(1-\frac{\lambda}{\mu}\right)^2}$$

$$= \frac{\frac{\lambda}{\mu}}{1-\frac{\lambda}{\mu}} = \frac{\lambda}{\mu-\lambda}$$

2. Expected no. of customers in the queue (average queue length)

This is denoted by N_q, where

$$N_q = E(q) = N_s - \frac{\lambda}{\mu} = \frac{\lambda}{\mu-\lambda} - \frac{\lambda}{\mu} = \frac{\lambda^2}{\mu(\mu-\lambda)}$$

3. Average waiting time in the system (in the queue and in the service)

- Let a customer remain in the system for the time T_s. During this time λT_s additional arrivals will occur;

$$\Rightarrow \quad E(N) = N_s = \lambda\cdot T_s$$

$\Rightarrow \quad \frac{\lambda}{\mu-\lambda} = \lambda T_s$, therefore the average waiting time in the system W_s is given by

$$W_s = T_s = \frac{1}{\mu-\lambda}$$

4. Average waiting time in the queue (average queueing time)

- The total time T_s consists of the average waiting time in the queue plus the expected time of receiving the service.
- Average time spent in service is given by $\frac{1}{\mu}$, i.e. the time taken to serve one customer.

$$T_q = T_s - \frac{1}{\mu}$$

$$\Rightarrow \quad W_q = T_q \quad \Rightarrow \quad \frac{1}{\mu-\lambda} - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu-\lambda)}$$

5. The probability that there are at least k customers in the system

$$\Pr(\text{at least } k \text{ customers}) = \sum_{n=k}^{\infty} P_n = \sum_{n=k}^{\infty}\left(1-\frac{\lambda}{\mu}\right)\left(\frac{\lambda}{\mu}\right)^k = \left(1-\frac{\lambda}{\mu}\right)\sum_{n=k}^{\infty}\left(\frac{\lambda}{\mu}\right)^k$$

$$= \left(1-\frac{\lambda}{\mu}\right)\left[\left(\frac{\lambda}{\mu}\right)^k + \left(\frac{\lambda}{\mu}\right)^{k+1} + \left(\frac{\lambda}{\mu}\right)^{k+2} + \cdots\right]$$

$$= \left(1-\frac{\lambda}{\mu}\right)\cdot\left(\frac{\lambda}{\mu}\right)^k\left[1+\frac{\lambda}{\mu}+\left(\frac{\lambda}{\mu}\right)^2+\cdots\right] = \left(\frac{\lambda}{\mu}\right)^k$$

10.8 Parameters associated with a finite queue

— outline notebook, p. 28 —

1. Expected no. of customers in the system

This is denoted by N_s, where

$$N_s = E(N) = \sum_{n=0}^{N} nP_n = \sum_{n=0}^{N} n\left(\frac{\lambda}{\mu}\right)^n P_0 = \sum_{n=0}^{N} n\left(\frac{\lambda}{\mu}\right)^n\cdot\frac{1-\frac{\lambda}{\mu}}{1-\left(\frac{\lambda}{\mu}\right)^{N+1}}$$

$$= \frac{1-\frac{\lambda}{\mu}}{1-\left(\frac{\lambda}{\mu}\right)^{N+1}} \cdot \sum_{n=0}^{N} n\left(\frac{\lambda}{\mu}\right)^{n} = \frac{1-\theta}{1-\theta^{N+1}} \cdot \sum_{n=0}^{N} n\,\theta^{n} \qquad \cdots ①$$

consider

$$\sum_{n=0}^{N} n\theta^{n} = \theta + 2\theta^{2} + 3\theta^{3} + 4\theta^{4} + \cdots + N\theta^{N} = \theta\left[1 + 2\theta + 3\theta^{2} + 4\theta^{3} + \cdots + N\theta^{N-1}\right]$$

and

$$\sum_{n=0}^{N} \theta^{n} = 1 + \theta + \theta^{2} + \cdots + \theta^{N} = \frac{1-\theta^{N+1}}{1-\theta} \qquad \cdots ②$$

We differentiate equation ② on both sides with respect to (w.r.t.) θ to get

$$\sum_{n=0}^{N} n\theta^{n-1} = \frac{(1-\theta)\left[-(N+1)\theta^{N}\right] + 1 - \theta^{N+1}}{(1-\theta)^{2}}$$

$$\Rightarrow \quad \sum_{n=0}^{N} n\theta^{n-1} = -\frac{(N+1)\theta^{N}}{1-\theta} + \frac{1-\theta^{N+1}}{(1-\theta)^{2}}$$

we introduce θ on both sides:

$$\sum_{n=0}^{N} n\theta^{n} = \theta\left[-\frac{(N+1)\theta^{N}}{1-\theta} + \frac{1-\theta^{N+1}}{(1-\theta)^{2}}\right]$$

from equation ① our N_s becomes

$$N_s = \frac{1-\theta}{1-\theta^{N+1}} \cdot \theta\left[-\frac{(N+1)\theta^{N}}{1-\theta} + \frac{1-\theta^{N+1}}{(1-\theta)^{2}}\right]$$

therefore

$$N_s = \frac{\theta}{1-\theta} - \frac{(N+1)\theta^{N+1}}{1-\theta^{N+1}} = \frac{\frac{\lambda}{\mu}}{1-\frac{\lambda}{\mu}} - \frac{(N+1)\left(\frac{\lambda}{\mu}\right)^{N+1}}{1-\left(\frac{\lambda}{\mu}\right)^{N+1}} = \frac{\lambda}{\mu-\lambda} - \frac{(N+1)\left(\frac{\lambda}{\mu}\right)^{N+1}}{1-\left(\frac{\lambda}{\mu}\right)^{N+1}}$$

2. Expected queue length

This is denoted by N_q:

$$N_q = \sum_{n=0}^{N} (n-1)P_n = \sum_{n=1}^{N} nP_n - \sum_{n=1}^{N} P_n = N_s - (1-P_0)$$

since N_s is given by the expression above.

- Note that we cannot compute W_q and W_s for a given λ because no raw arrivals are allowed in the system once the system is full; but if λ is replaced by $\lambda' = \lambda(1-P_N)$, which is the effective rate of arrivals, then

$$W_s = \frac{N_s}{\lambda'} \qquad \text{and} \qquad W_q = W_s - \frac{1}{\mu} = \frac{N_s}{\lambda(1-P_N)} - \frac{1}{\mu}$$

10.9 Worked examples

Example 1 (car-wash facility)

— outline notebook, p. 27 —

In a car-wash service facility, the information gathered indicates that cars arrive for service in a Poisson distribution with mean 5 cars per hour. The time for washing and cleaning each car varies but is found to follow an exponential distribution with 10 mins/car. The facility cannot handle more than one car at a time. Assuming that the calling source is infinite, find:

i. The no. of parking spaces that should be available for the cars arriving at the facility.

ii. The parking spaces that should be provided so that an arriving car can park at least 80% of the time.

iii. The % of time the facility is idle.

iv. The expected waiting time from the moment the car arrives until it leaves the facility.

Soln. Arrival rate $\lambda = 5$ cars per hour; departure rate $\mu = 6$ cars per hour. Clearly $\lambda < \mu$ and so the system can operate under steady-state conditions.

i) The expected no. of cars in the queue:

$$N_q = \frac{\lambda^2}{\mu(\mu-\lambda)} = \frac{5^2}{6(6-5)} = \frac{25}{6} = 4.1667 \approx \text{about } 5 \text{ parking spaces}$$

ii) This requirement can be expressed as a probability statement:

$$P_0 + P_1 + P_2 + P_3 + P_4 + P_5 \geq 0.8,$$

where S (finite) is the no. of parking spaces we want to determine. But $P_n = \theta^n(1-\theta)$:

$$\Rightarrow \quad 1 - \theta + \theta(1-\theta) + \theta^2(1-\theta) + \cdots + \theta^S(1-\theta) \geq 0.8$$

$$1 - \theta\left[1 - \theta + \theta^2 + \cdots + \theta^S\right] \geq 0.8$$

$$1 - \theta \cdot \frac{1-\theta^{S+1}}{1-\theta} \geq 0.8 \quad \Rightarrow \quad 1 - \theta^{S+1} \geq 0.8$$

$$\Rightarrow \quad \theta^{S+1} \leq 0.2$$

$\left(\frac{5}{6}\right)^{S+1} \leq 0.2$, we take logs on both sides and solve for S to get $(S = 7.8)$

$$S + 1 = \frac{\log(0.2)}{\log(0.8)} = \frac{-0.69897}{-0.079}$$

$$S + 1 = 8.84$$

$$S = 8.84 - 1 = 7.8 \qquad \text{(about 8 parking spaces)}$$

iii) The % of time the facility is idle is equivalent to P_0:

$$\Rightarrow \quad P_0 = 1 - \theta = 1 - \frac{5}{6} = \frac{1}{6} = 17\% \text{ of the time the facility is idle}$$

iv) We need W_s:

$$W_s = \frac{1}{\mu - \lambda} = \frac{1}{6-5} = 1 \text{ hr}$$

Example 2 (railway station)

— outline notebook, pp. 28–29 —

At a railway station, only one train is handled at a time. The railway yard is sufficient only for 2 trains to wait while others are given a signal to leave the station. Trains arrive at an average of 6 per hour and the railway station can handle them at an average of 12/hr. Assuming Poisson arrivals and exponential service time distribution, find:

a. The steady-state probabilities P_0 and P_N.

b. The average waiting time of a new train coming into the system.

Soln. $\lambda = 6, \mu = 12$; $\lambda < \mu$ meaning the system can operate under steady-state conditions. Now $N = 2$.

a)

$$P_0 = \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^{N+1}} = \frac{1 - \frac{6}{12}}{1 - \left(\frac{6}{12}\right)^3} = \frac{0.5}{1 - 0.125} = \frac{0.5}{0.875} = \frac{4}{7}$$

$$P_N = \left(\frac{6}{12}\right)^2 P_0 = (0.5)^2 \cdot \frac{4}{7} = \frac{1}{7}$$

b)

$$W_s = \frac{N_s}{\lambda(1 - P_N)}$$

but

$$N_s = \frac{\lambda}{\mu - \lambda} - \frac{(N+1)\left(\frac{\lambda}{\mu}\right)^{N+1}}{1 - \left(\frac{\lambda}{\mu}\right)^{N+1}} = \frac{4}{7}$$

$$W_s = \frac{\frac{4}{7}}{6\left(1-\frac{1}{7}\right)} = \frac{1}{9}$$

— outline notebook, p. 29 (dated Wednesday 16th April 2025) —

11. Stochastic Processes with Discrete States in Discrete Time

- Here we consider a class of stochastic processes whereby the time is discrete and the state space is also discrete.
- A good example is a Markov chain.

11.1 Markov chain

- A Markov chain is a stochastic process (a Markov process) whose state space is discrete in discrete time.

Illustration

Consider an experiment of tossing a coin several times (n times) where the possible outcome(s) in each trial is head or tails with probability p and q (respectively).

- Let the head denote 1 and the tail denote 0, and the random variable denoting the result of the nth trial be X_n.
- For $n = 0, 1, 2, 3, \ldots, n$: $\Pr(X_n = 1) = p$ and $\Pr(X_n = 0) = q$; thus the sequence of random variables $X_1, X_2, \ldots, X_n$ are independent and the results of the nth trial are $1, 0, \ldots, (n-1)$.
- The sum $S_n = X_1 + X_2 + X_3 + \cdots + X_n$ gives the no. of heads (successes) in the first n trials and its possible outcomes are $0, 1, 2, 3, \ldots, n$.
- We realise that

$$S_{n+1} = S_n + X_{n+1}$$

Now given that $S_n = j, j = 0, 1, 2, 3, \ldots, n$, the random variable S_{n+1} can assume only 2 possible values, i.e.

$S_{n+1} = j$, with probability q and $S_{n+1} = j+1$, with probability p

— outline notebook, p. 30 —

These probabilities are not affected by the values $S_1, S_2, \ldots, S_{n-1}$, thus

$$\Pr\left(S_{n+1} = j / S_n = j\right) = q, \qquad \Pr\left(S_{n+1} = j+1 / S_n = j\right) = p$$

This is an example of a Markov chain.

- Markov chains are applicable in dynamic systems, for example:
  1. In analysing inventory problems, population growth problems, accounting problems, etc.

11.2 Multiplication of probabilities

- Let E_j and E_k be two events, then

$$\Pr\left[E_j E_k\right] = \Pr\left[E_k / E_j\right] \cdot \Pr\left[E_j\right] = \Pr\left[E_j\right] \cdot \Pr\left[E_k / E_j\right] = g_j \cdot p_{jk}$$

where $g_j = \Pr[E_j]$ and $p_{jk} = \Pr[E_k / E_j]$.

Again,

$$\Pr\left[E_j E_k E_r\right] = \Pr\left[E_k E_r / E_j\right] \cdot \Pr\left[E_j\right] = \Pr\left[E_r / E_j E_k\right] \cdot \Pr\left[E_k / E_j\right] \cdot \Pr\left[E_j\right] = p_{kr} \cdot p_{jk} \cdot g_j$$

11.3 Definition of a Markov chain

A sequence of trials with possible outcomes $E_{j_0}, E_{j_1}, E_{j_2}, \ldots, E_{j_n}$ is called a Markov chain if the probability of

$$\Pr\left[E_{j_0}, E_{j_1}, \ldots, E_{j_{n-1}}, E_{j_n}\right] = g_{j_0} \cdot p_{j_0 j_1} \cdot p_{j_1 j_2} \cdots p_{j_{n-1} j_n},$$

where $g_{j_0} = \Pr[E_{j_0}]$ is called the probability at the initial trial and p_{jk} is the fixed probability of event E_k given event E_j.

- We shall call the event E_j the state E_j or simply **state** j.
- The conditional probability p_{jk} shall be called the **transitional probability** from state E_j to state E_k, i.e.

$$\Pr\left[E_k/E_j\right] = \Pr\left[E_j \to E_k\right] = \Pr(\text{moving from state } j \text{ to state } k)$$

11.4 The transition matrix

We can express the transitional probabilities in a matrix form as follows:

$$P = \begin{pmatrix} p_{00} & p_{01} & p_{02} & p_{03} & \cdots \\ p_{10} & p_{11} & p_{12} & p_{13} & \cdots \\ p_{20} & p_{21} & p_{22} & p_{23} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \end{pmatrix}$$

where the row index is the present state E_j and the column index is the next state E_k (the notebook labels the rows E_1, E_2, E_3 and the columns E_1, E_2, E_3).

e.g.

$$p_{23} = \Pr\left[E_2 \to E_3\right], \qquad p_{12} = \Pr\left[E_1 \to E_2\right]$$

- Note that the transitional matrix is either finite or infinite.
- Any matrix whose elements $p_{jk} \geq 0$ and $\sum_k p_{jk} = 1$ is called a **stochastic matrix**.

e.g.

$$P = \begin{pmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} \end{pmatrix} \quad \to \quad \text{double stochastic matrix}$$

$$\sum \text{rows} = 1, \qquad \sum \text{columns} = 1$$

11.5 Higher-order transitional probabilities

- $p_{jk}^{(n)}$ is the probability of moving from state E_j to state E_k in n steps. In particular, $p_{jk}^{(2)}$ is the probability of moving from state E_j to state E_k in 2 steps, i.e.

(the notebook continues on the following page, which is not part of the scanned set.)

— outline notebook, p. 15 —

Appendix A. Past Examination Questions (Last Year's CAT)

The notebook records the following class-test questions with their worked answers.

a) Define the following terms:

i) Discrete-time stochastic process: If T is discrete, e.g. $T = \{0, 1, 2, \ldots\}$, then the process $\{X_t\}_{t \in T}$ is called a discrete-time stochastic process.

ii) Continuous-time stochastic process: If T is continuous (in an interval of $\mathbb{R}$, $T = [0, \infty)$), then the process $\{X_t\}_{t \in T}$ is called a continuous-time stochastic process.

b) Let X be a discrete random variable with probability generating function

$$P(s) = \frac{\theta s^3}{1 - (1 - \theta)s}$$

Find the distribution of X.

Solution. Given $P(X = k)$ where $k = \{0, 1, 2, \ldots\}$ is discrete,

$$P(s) = \sum_{k=0}^{\infty} p_k s^k \qquad \text{where } p_k = \Pr(X = k)$$

Therefore

$$P(s) = \sum_{k=0}^{\infty} \Pr(X = k)s^k = E\left[s^X\right], \qquad E[x] = P^{'}(s)$$

$$P(s) = \frac{1}{1-(1-\theta)s} = \sum_{n=0}^{\infty} (1-\theta)^n s^n \qquad \times \theta s^3$$

$$P(s) = \theta s^3 \sum_{n=0}^{\infty} (1-\theta)^n s^n = \sum_{n=0}^{\infty} \theta(1-\theta)^n s^{n+3}$$

where $n + 3 = k$, i.e. $n = k - 3$:

$$P(s) = \sum_{k=0}^{\infty} \theta(1-\theta)^{k-3} s^k \quad \Rightarrow \quad \Pr(X = k)$$

$$P(X = k) = \theta(1-\theta)^{k-3}, \qquad \text{for } k \geq 3$$

c) Let X be a random variable such that $\Pr(X = k) = p_k$ and $\Pr(X \leq k) = q_k = p_0 + p_1 + \cdots + p_k$ for $k \geq 0$. Obtain the generating function of q_k in terms of the p.g.f. of p_k.

Solution.

$$\Pr(X = k) = p_k$$

The cumulative probability function is

$$P(X \leq k) = q_k = p_k + p_{k-1} + \cdots + p_1 + p_0, \qquad \text{for } k \geq 0$$

$$p_k = q_k - q_{k-1}$$

$$q_{k-1} = p_{k-1} + p_{k-2} + p_{k-3} + \cdots + p_1 + p_0$$

$$q_k = q_{k-1} + p_k, \qquad k \geq 1$$

Recall

$$P(s) = \sum_{k=0}^{\infty} p_k s^k \qquad \text{and} \qquad Q(s) = \sum_{k=0}^{\infty} q_k s^k$$

So

$$\sum_{k=0}^{\infty} p_k s^k = \sum_{k=0}^{\infty} q_k s^k - \sum_{k=1}^{\infty} q_{k-1} s^k$$

$$P(s) - P(0) = Q(s) + q_0 - s \sum_{k=1}^{\infty} q_{k-1} s^{k-1}$$

$$P(s) - p_0 = Q(s) - q_0 - sQ(s)$$

$$Q(s)(1-s) = P(s) - (p_0 - q_0) \qquad \text{since } p_0 = q_0$$

$$Q(s) = \frac{P(s) - 0}{1-s} = \frac{P(s)}{1-s}, \qquad |s| < 1$$

d) If $X = x_1 + x_2$, then $W = \text{Pgf}(x_1) \cdot \text{Pgf}(x_2)$ for independent random variables of cases x_1 and x_2.

$$X \sim \text{Bin}(4, \theta) \quad \text{and} \quad Y \sim \text{Bin}(3, \theta) \qquad \text{(binomial distribution)}$$

P.g.f. of the binomial distribution $= (q + ps)^n$:

$$X = \big((1-\theta) + \theta s\big)^4 = \big(\theta s + (1-\theta)\big)^4$$

$$Y = \big((1-\theta) + \theta s\big)^3 = \big(\theta s + (1-\theta)\big)^3$$

$$W = \big(\theta s + (1-\theta)\big)^4 \cdot \big(\theta s + (1-\theta)\big)^3 = \big(\theta s + (1-\theta)\big)^7$$

$$W = \big((1-\theta) + \theta s\big)^7 = W \sim \text{Bin}(7, \theta)$$

with $q = 1 - p$.

Appendix B. Notes on the Originals

The following are minor slips or ambiguities in the handwritten originals. They are transcribed in the text as written; the corrected or clarified forms are given here.

1. **Poisson process, equation (X).** On outline-notebook p. 8 the equation is quoted as $P_n^{'}(t) = \lambda P_n(t) + \lambda P_{n-1}(t)$; the sign of the first term is minus, as derived on p. 7: $P_n^{'}(t) = -\lambda P_n(t) + \lambda P_{n-1}(t)$.

2. **Birth–death d.d.e., page 18.** The equation is quoted as $P_n'(t) = n(\lambda+\mu)P_n(t) + (n-1)\lambda P_{n-1}(t) + (n+1)\mu P_{n+1}(t)$; as derived on p. 11 the first term carries a minus sign. The derivation that follows treats it correctly.
3. **Queuing table, page 23.** In the scan the column entries sit half a row below their labels, and the fourth event is written as a repeat of the third. The probability statement that follows uses exactly the four probabilities (i)–(iv).
4. **Queuing table, page 24.** The note "$1 - \mu\Delta t = 1$ since $\mu\Delta t = 0$ as $\mu = 0$" mixes μ (the departure rate) with the case $n = 0$, where no customer can depart; the algebraic effect is the same — the factor is 1.
5. **Geometric distribution (outline notebook p. 16).** The expansion there starts at $k = 1$, i.e. $P(s) = \frac{ps}{1-qs}$; the extended set and lecture notes start at $k = 0$. Both are shown in §5.4.
6. **Bernoulli p.g.f. (extended set, p. 4).** The line introducing $P(s) = q + ps$ says "The p.d.f. of x is given by"; it should read p.g.f.
7. **Geometric variance (lecture notes, p. 6).** The notes write $\text{Var}(x) = P''(1) + P'(1) + \left[P'(1)\right]^2$; the last sign should be minus, as in the extended set and in §5.4.
8. **Bivariate variances (lecture notes, p. 14).** The first moment term in $\sigma_x^2 = E[X(X-1)] + E(x)^2 - \left[E(x)\right]^2$ should be $E(x)$.
9. **Convergence interval of the g.f.** The outline notebook and extended set write $-s_0 < s < s_0$; the lecture notes write $s_0 < s < s_1$. Both are recorded in §3.1.
10. **Zero growth process, mean.** The notebook states $E(N) = G'(1,t) = 1$ "to show"; differentiating $G(s,t) = \frac{\lambda t + (1-\lambda t)s}{1 + \lambda t - \lambda t s}$ at $s = 1$ gives 1, and $\text{Var}(N) = G''(1,t) + G'(1,t) - \left(G'(1,t)\right)^2 = 2\lambda t$, the two results left as exercises.

Appendix C. Coverage Map (source pages → sections)

Section	Outline notebook (30 pp.)	Extended set (7 pp.)	Lecture notes (15 pp.)
1. Course outline, description, outcomes	p. 1	—	—
2. Random phenomena; stochastic process definition	p. 1	p. 1	p. 1
2.3 Time, states, state space	p. 1	—	—
2.4 Examples of stochastic processes	p. 1	p. 1	p. 1
2.5 Common stochastic processes	p. 1	—	—
3.1 Definition of the g.f.; convergence	pp. 1–2	pp. 1–2	p. 2
3.2 Coefficient extraction by differentiation	p. 2	—	p. 2
3.3 Examples 1–4	p. 2	p. 2	pp. 2–3
3.3 Example 5 (recurrence)	—	p. 2	pp. 3–4
4.1 p.g.f. definition; $E[s^X]$	pp. 2–3	p. 3	p. 4
4.2 Sequence that is not a p.g.f.	p. 2	p. 3	p. 4
4.3 Mean and variance from the p.g.f.	p. 3	pp. 3–4	pp. 4–5
4.4 m.g.f. link; integral; $Y = mX + n$; recovering p_k	—	pp. 6–7	pp. 7–8
5.1 Bernoulli	pp. 3–4	p. 4	p. 5
5.2 Binomial	pp. 3–4	pp. 4–5	pp. 5–6
5.3 Poisson	p. 4	p. 6	p. 6 ("Read")

Section	Outline notebook (30 pp.)	Extended set (7 pp.)	Lecture notes (15 pp.)
5.4 Geometric	p. 4 (exercise), p. 16 (working)	p. 5	p. 6
6.1 Upper tail	pp. 4–5	—	pp. 10–11
6.2 Lower tail	p. 5	—	p. 12
7.1 Fixed number of terms; Poisson and Bernoulli sums	pp. 5–6	p. 7	pp. 8–9
7.2 Random number of terms; compound Poisson	p. 6	—	pp. 9–10
8. Poisson process: definition, d.d.e., solution	pp. 6–9	—	—
9.1–9.2 Birth–death process, d.d.e.	pp. 10–11	—	—
9.3 Pure birth process; special cases	p. 11	—	—
9.4 Yule–Furry process, solution	pp. 11–12	—	—
9.4a Second pass at the pure birth solution	pp. 16–17	—	—
9.5 Linear birth–death process	pp. 13–14	—	—
9.6 Zero growth process	pp. 20–21	—	—
9.7 Feller's method: mean and variance	pp. 18–19	—	—
10.1–10.3 Queuing models: characteristics, notation	pp. 21–22	—	—
10.4–10.6 Queuing as a birth–death process; steady state	pp. 23–25	—	—
10.7–10.8 Parameters, infinite and finite queue	pp. 25–26, 28	—	—
10.9 Worked examples	pp. 27–29	—	—
11. Markov chains	pp. 29–30	—	—
Appendix A. Past CAT questions	p. 15	—	—

Every page of all three notebooks is accounted for: outline notebook pages 1–30, extended set pages 1–7 and lecture notes pages 1–15.